

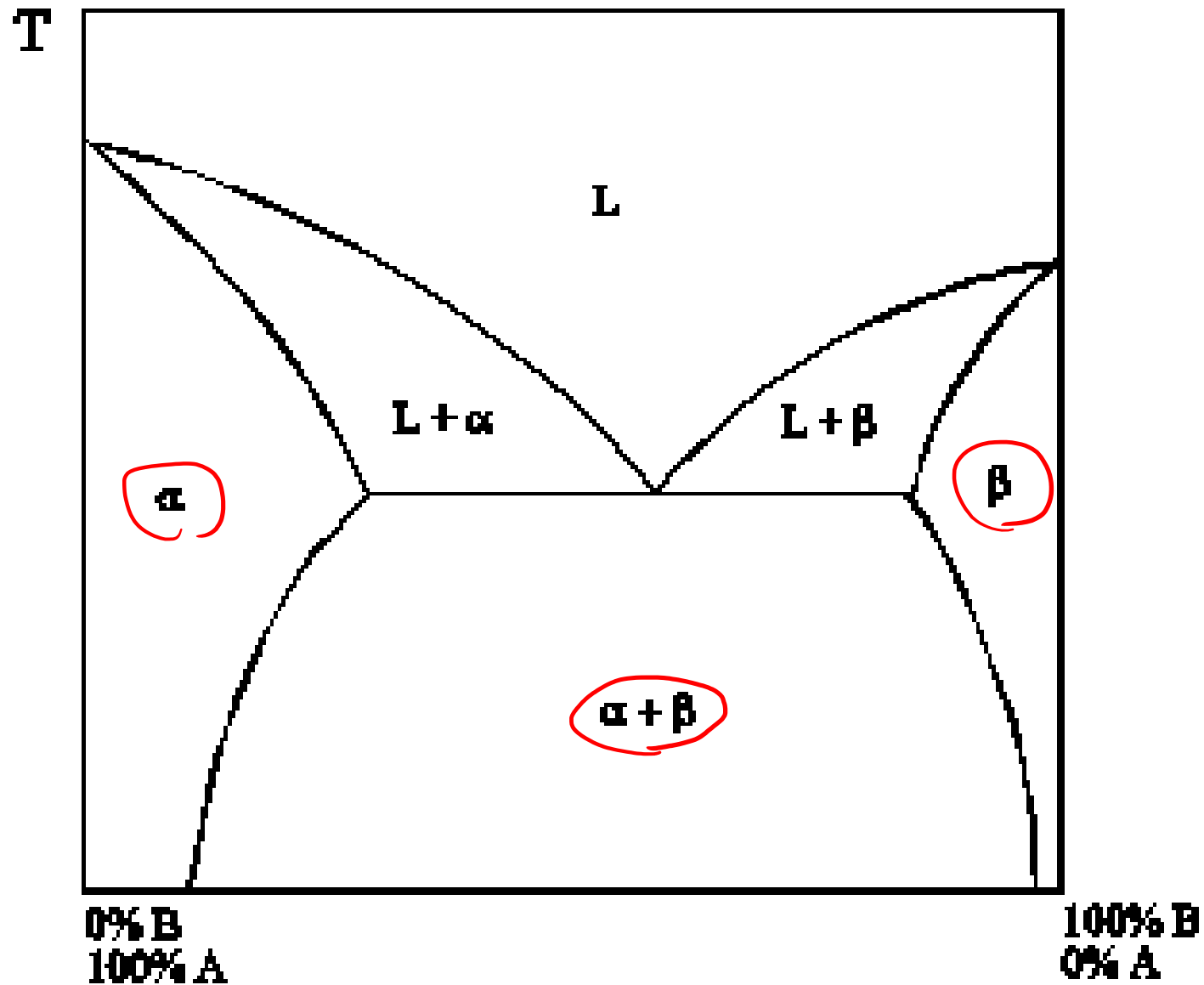
A



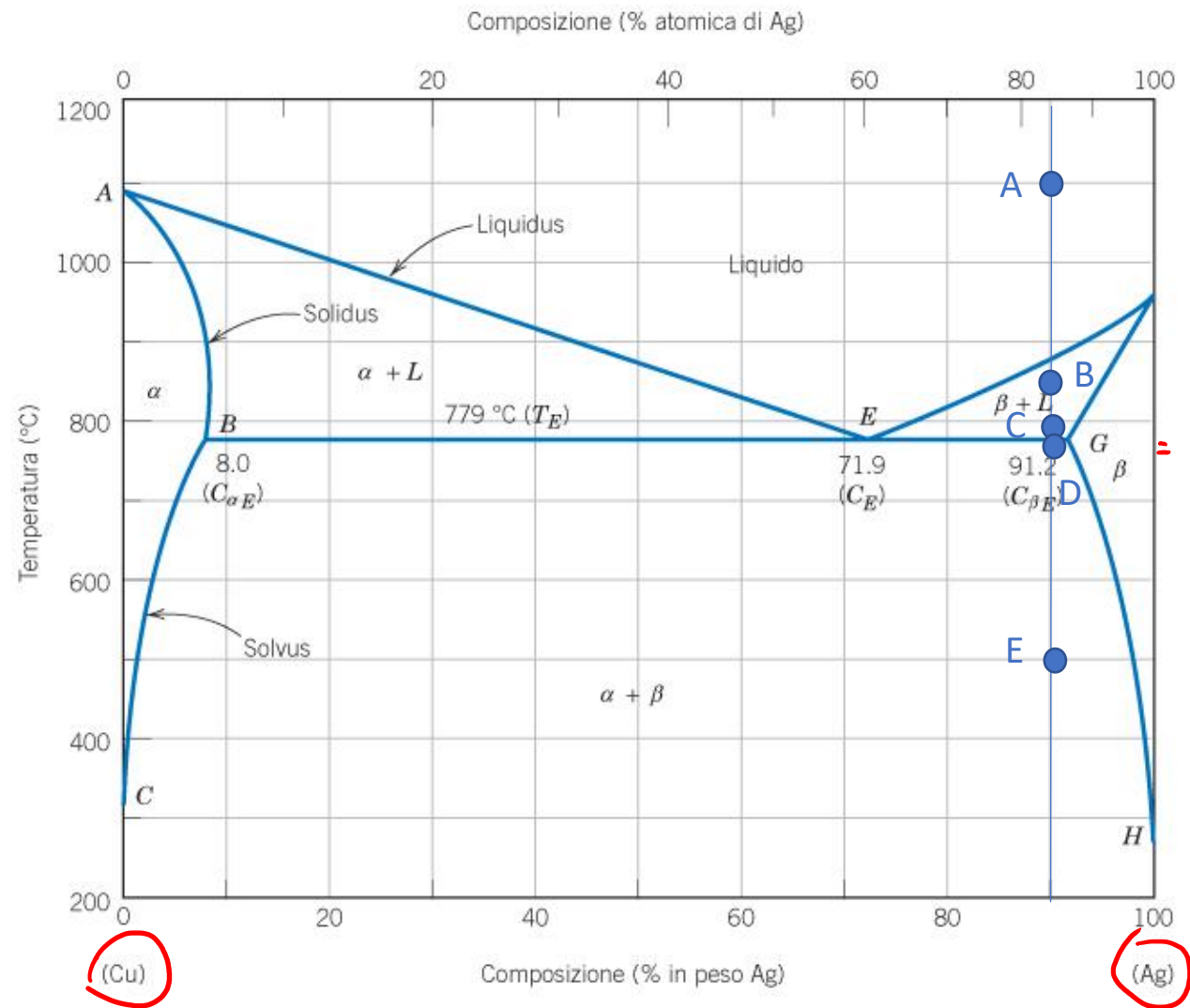
STM Exercise 7

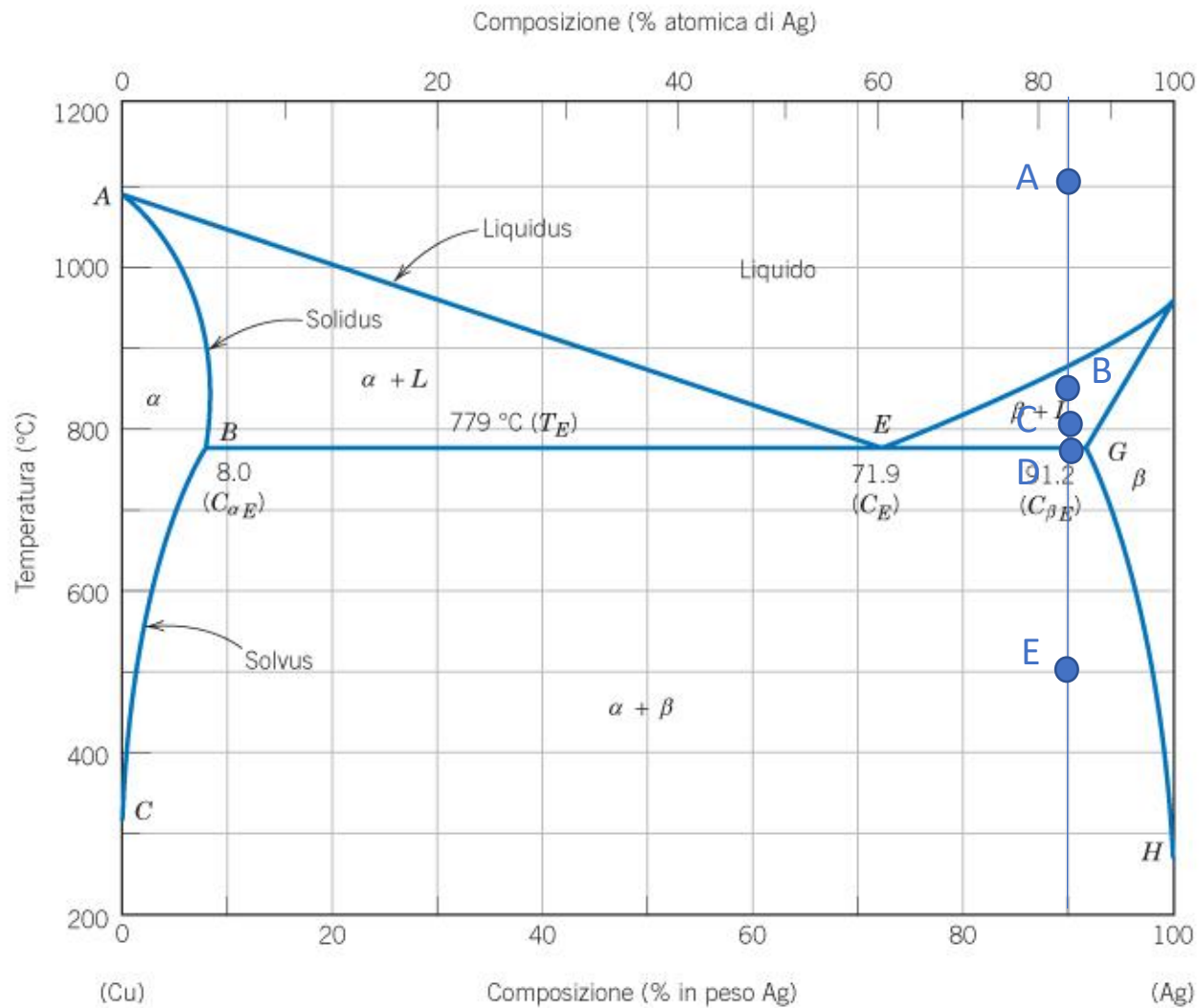
Phase Diagrams
(partial solubility in the solid state)

Complete solubility in the liquid state,
partial solubility in the solid state



Exercise 1: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.

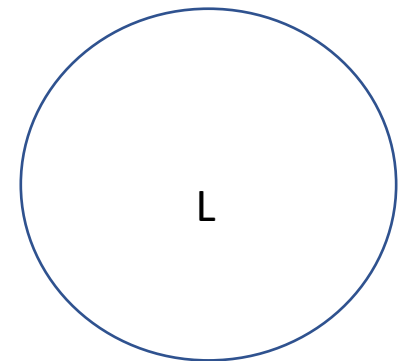


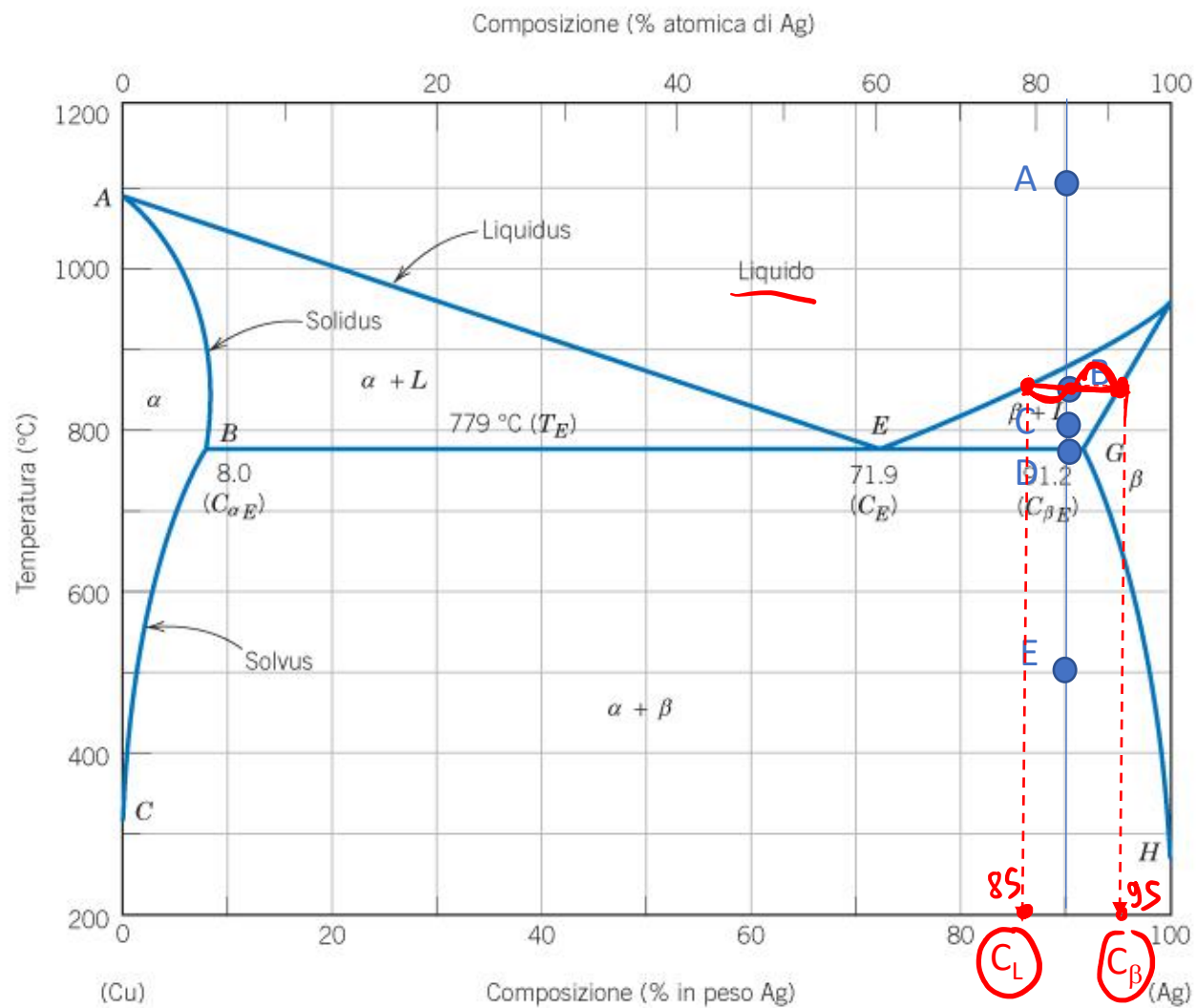


① C $\begin{cases} 90\% \text{ Ag} \\ 10\% \text{ Cu} \end{cases}$

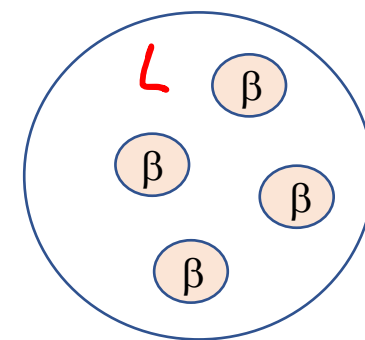
$P_h \rightarrow L$

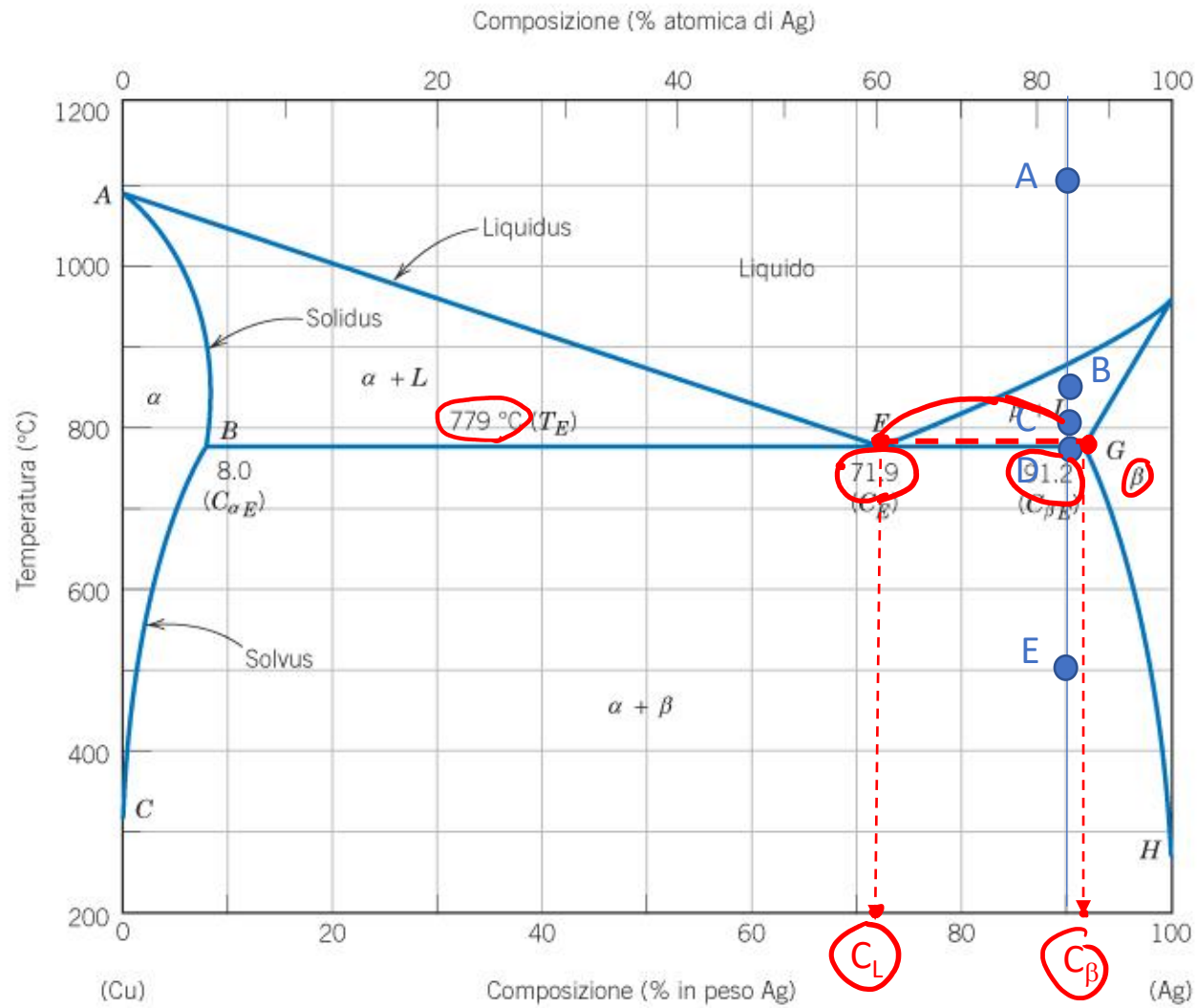
$\%_A \rightarrow 100\% \text{ L}$



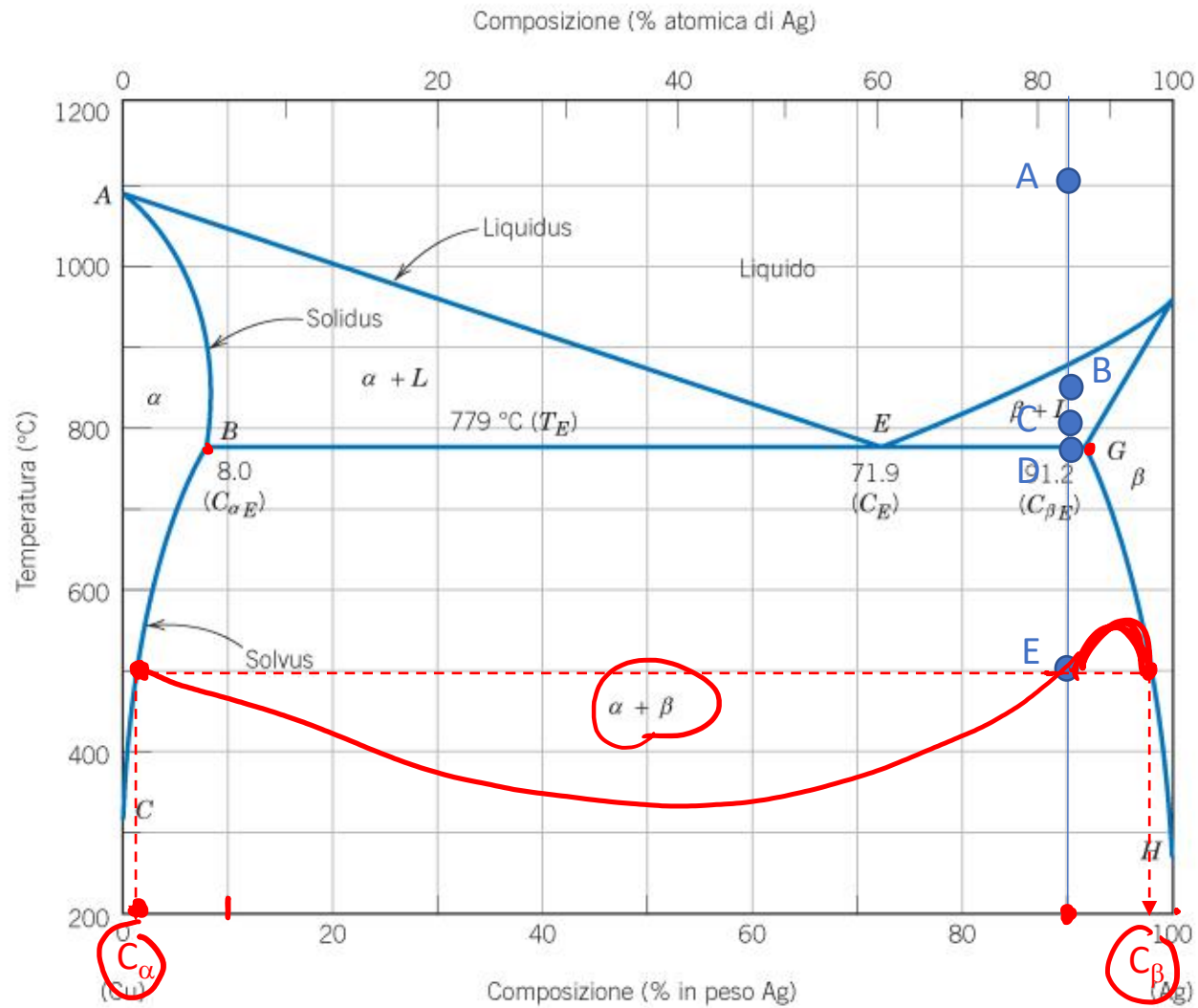


③	C	%
L	85% Ag 15% Cu	$\frac{95-90}{95-85} \cdot 100 = 50\%$
β	95% Ag 5% Cu	50%

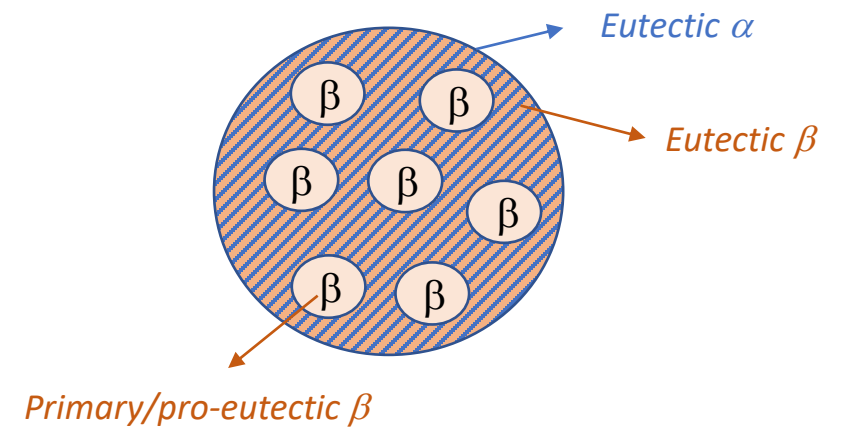




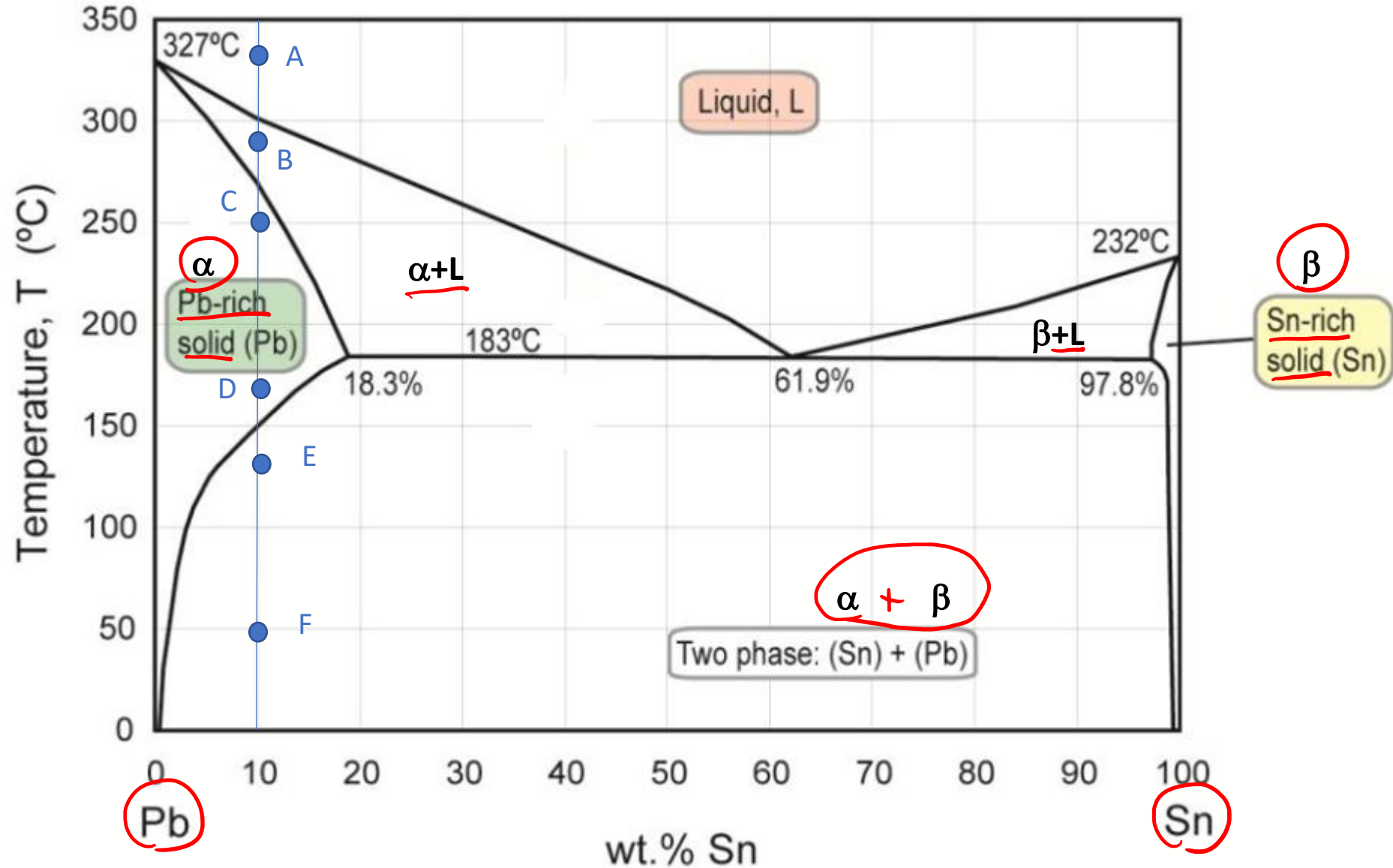
C	C	%
L	71,9% Ag 28,1% Cu	6,2%
β	91,2% Ag 8,8% Cu	$\frac{90 - 71,9}{91,2 - 71,9} \cdot 100 =$ 93,8%

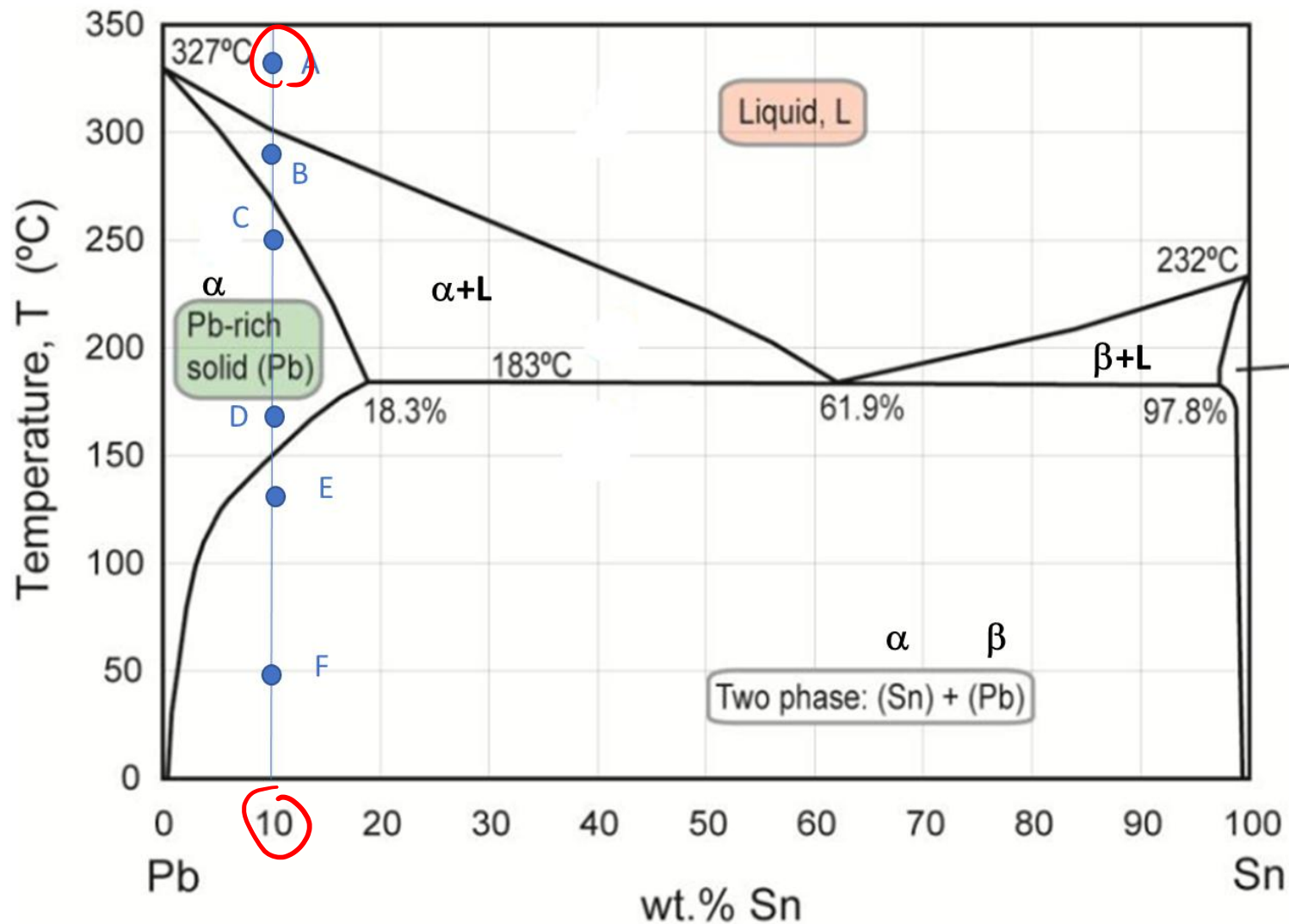


E	C	%.
α	1% Ag 99% Cu	$\frac{98 - 90}{98 - 1} \cdot 100 = 82\%$
β	98% Ag 2% Cu	91,8%



Exercise 2: determine the phase present, their composition and relative amounts in the points A, B, C, D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



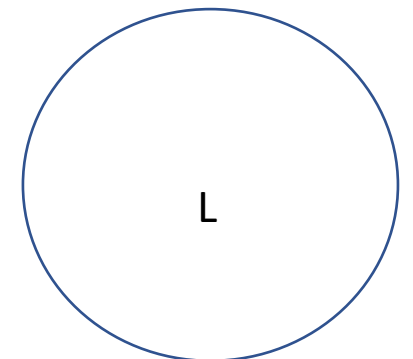


Point A:

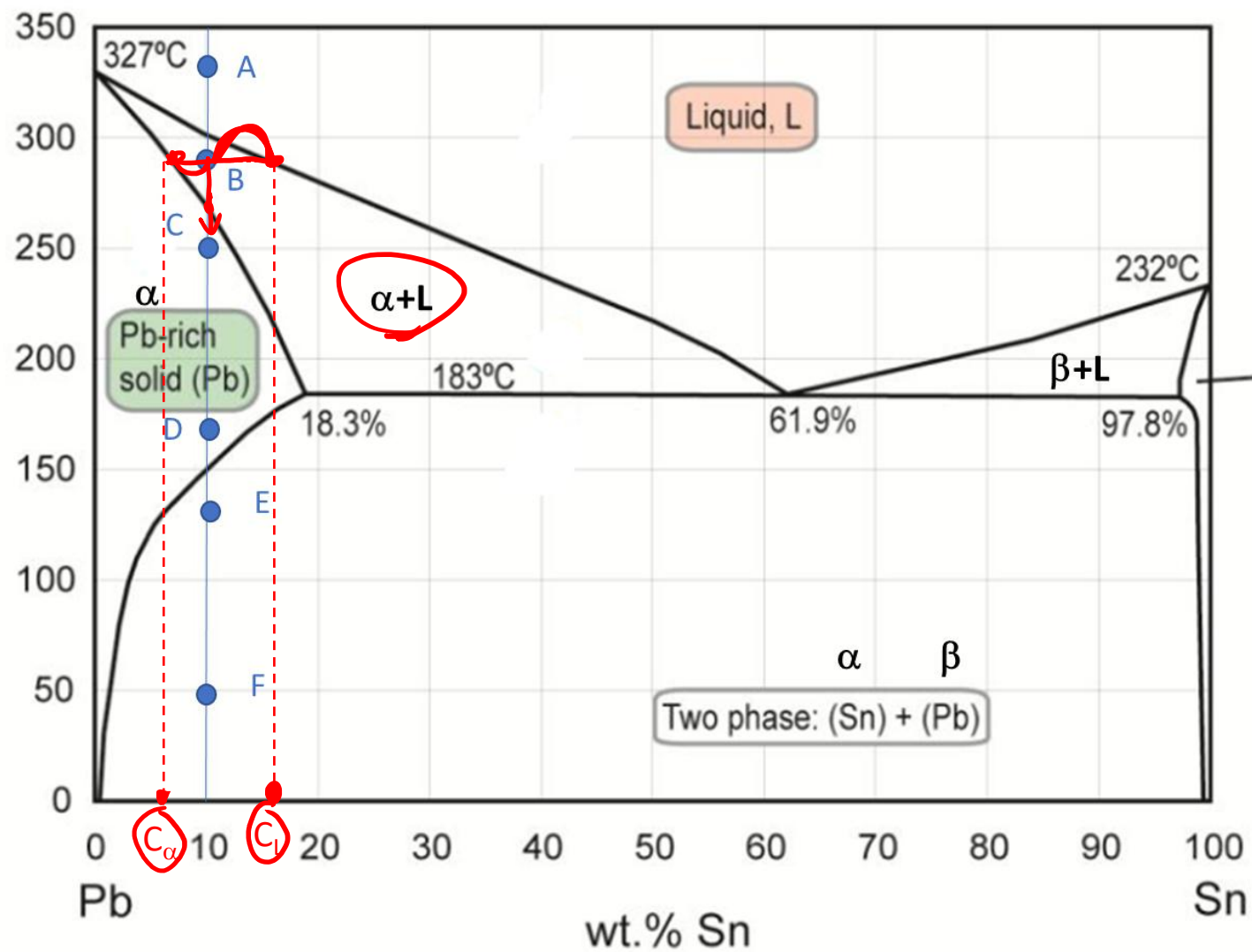
Phases: 1 phase liquid

Composition: 10% Sn, 90% Pb

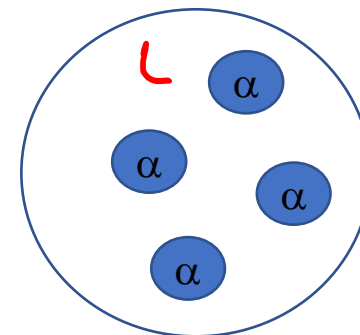
Relative amount: 100% liquid phase

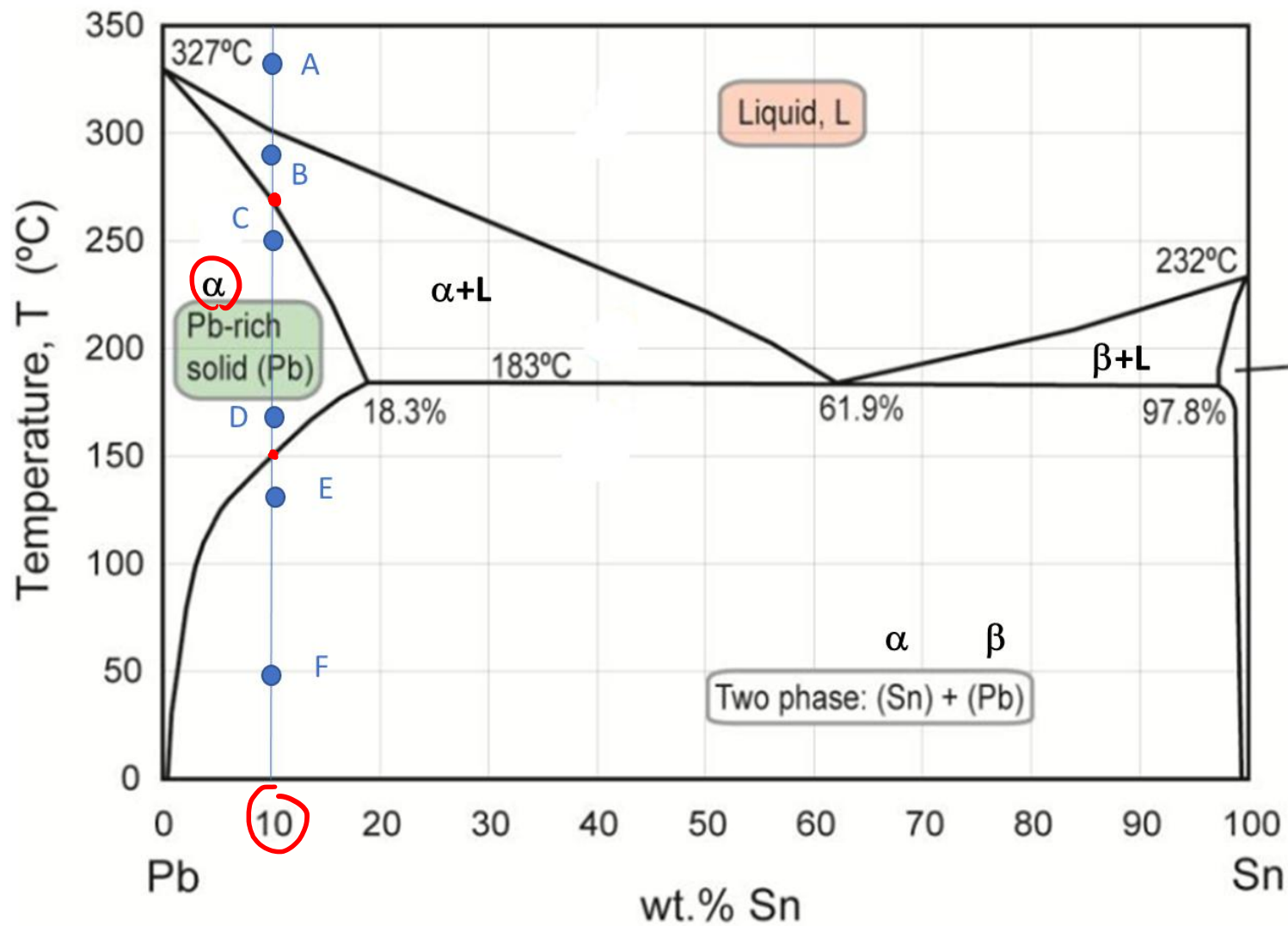


Temperature, T (°C)



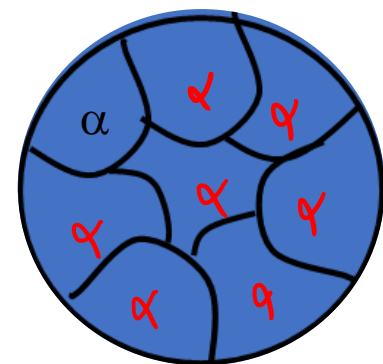
③	C	%
- L	15% Sn 85% Pb	28.6%
- α	8% Sn 92% Pb	$\frac{15-10}{15-8} \cdot 100 = 71.4\%$

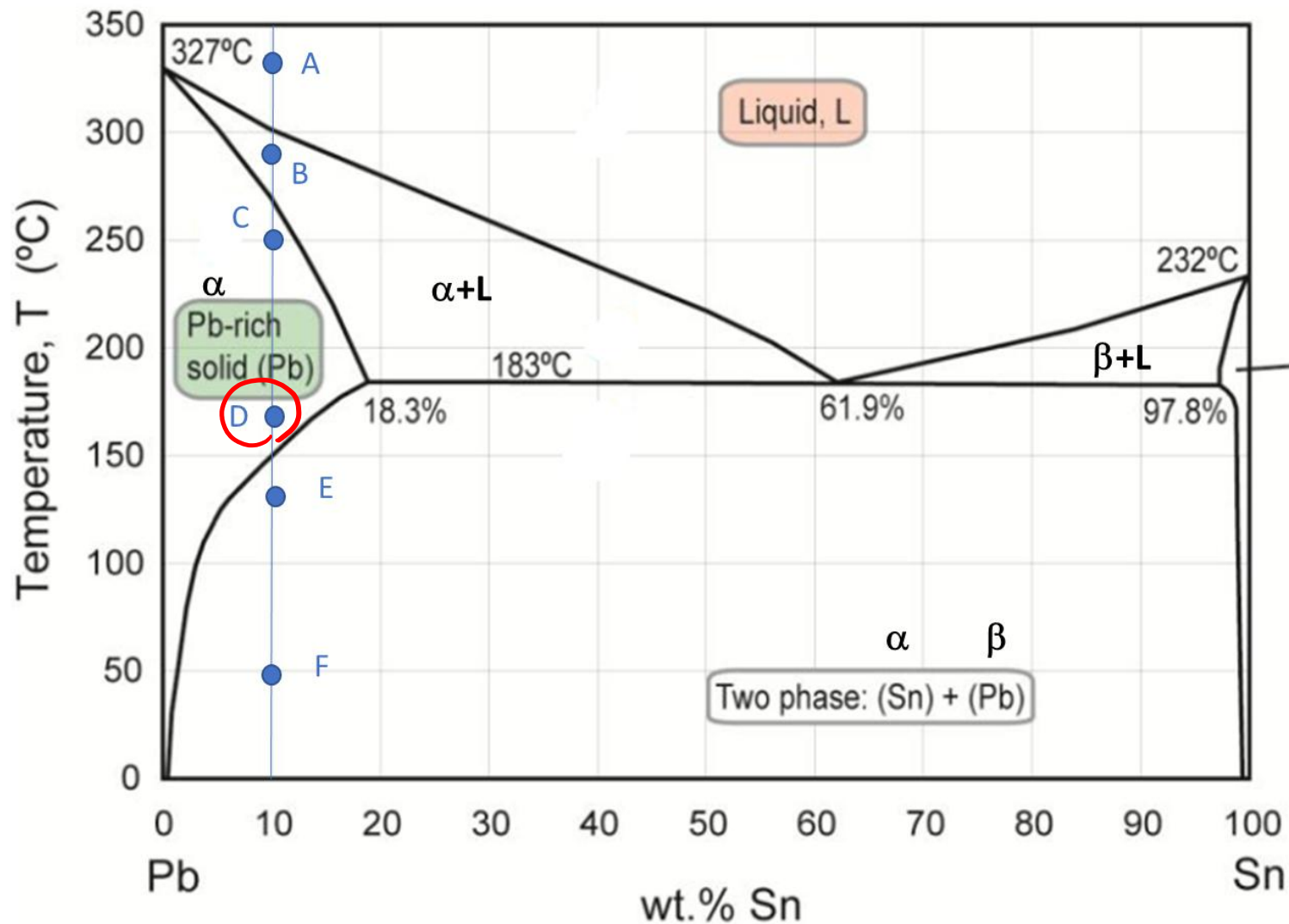




①	C	%
α	10% Sn 90% Pb	100%

↑



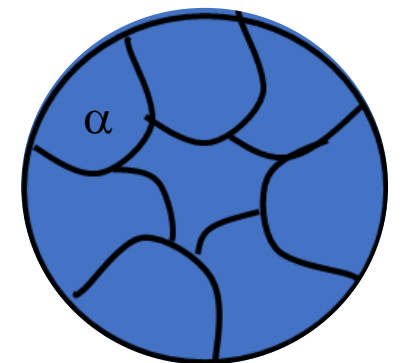


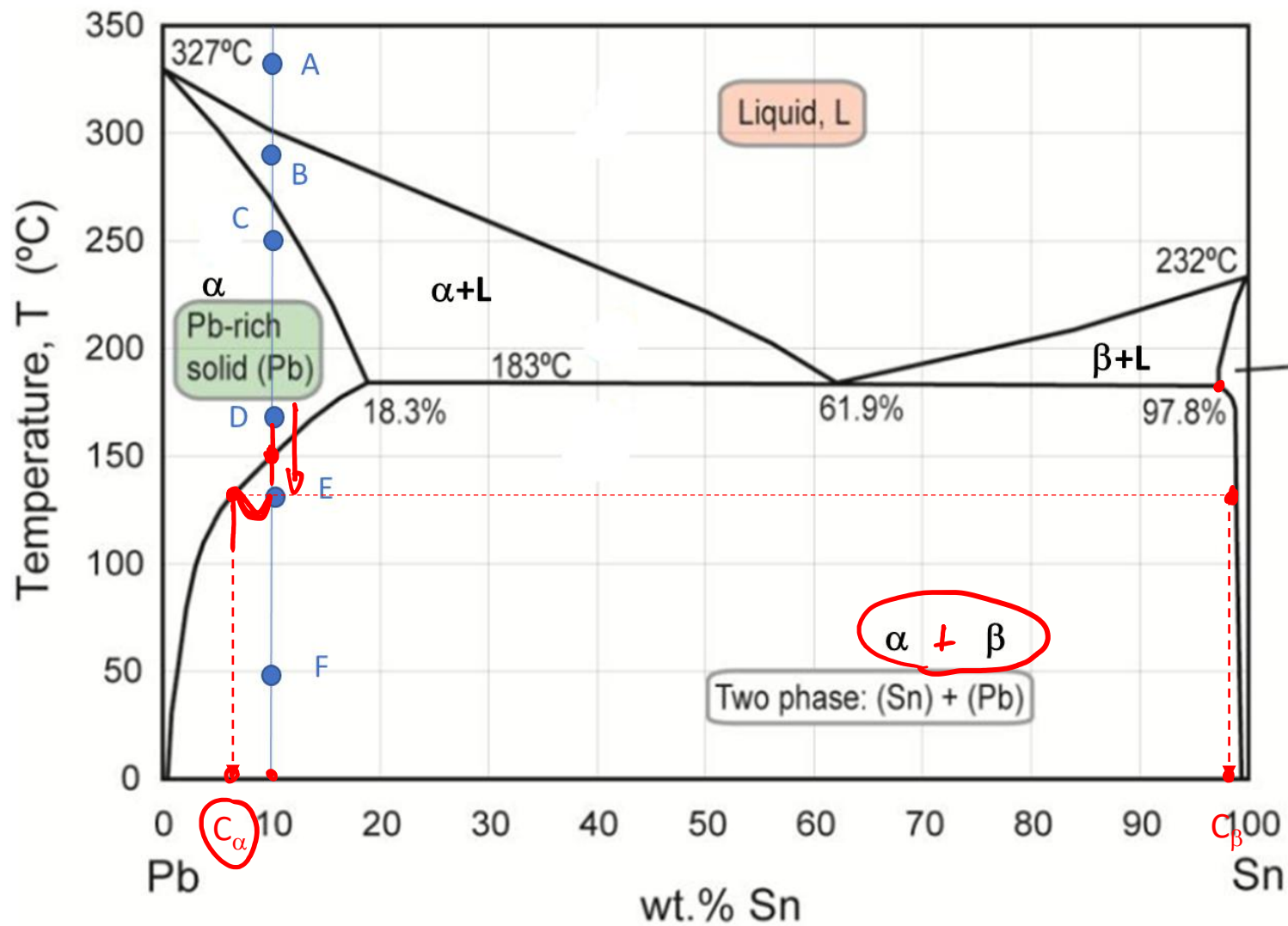
Point D:

Phases: 1 phase solid (α)

Composition: 10% Sn, 90% Pb

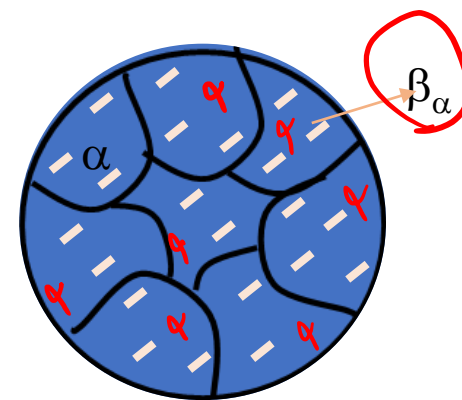
Relative amount: 100% solid phase α

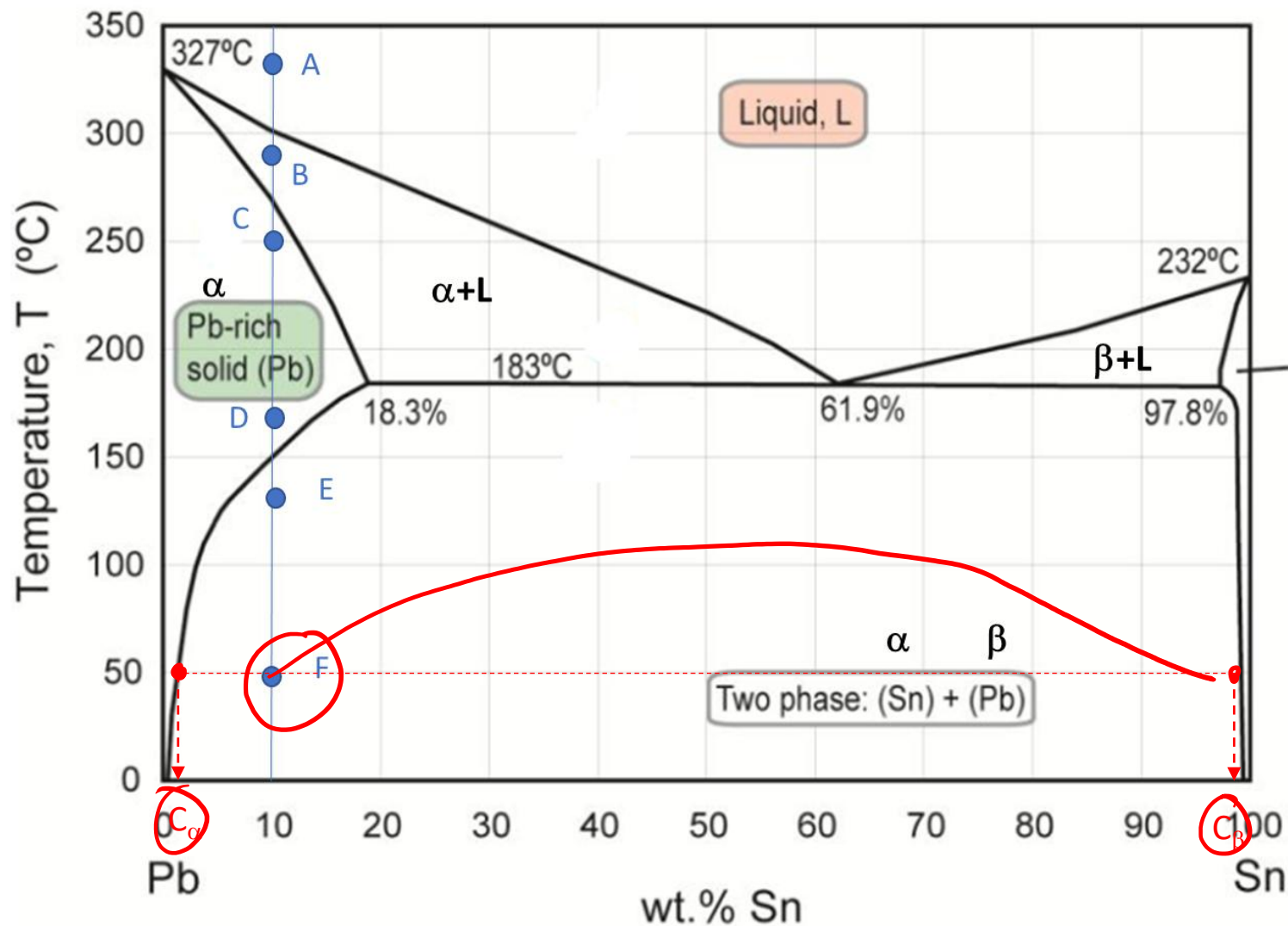




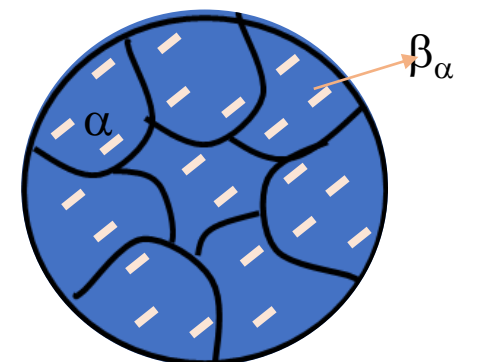
(E)	C	%
α	7% Sn 93% Pb	97%
β	99% Sn 1% Pb	3%

$\frac{10 - 7}{99 - 7} \cdot 100 =$

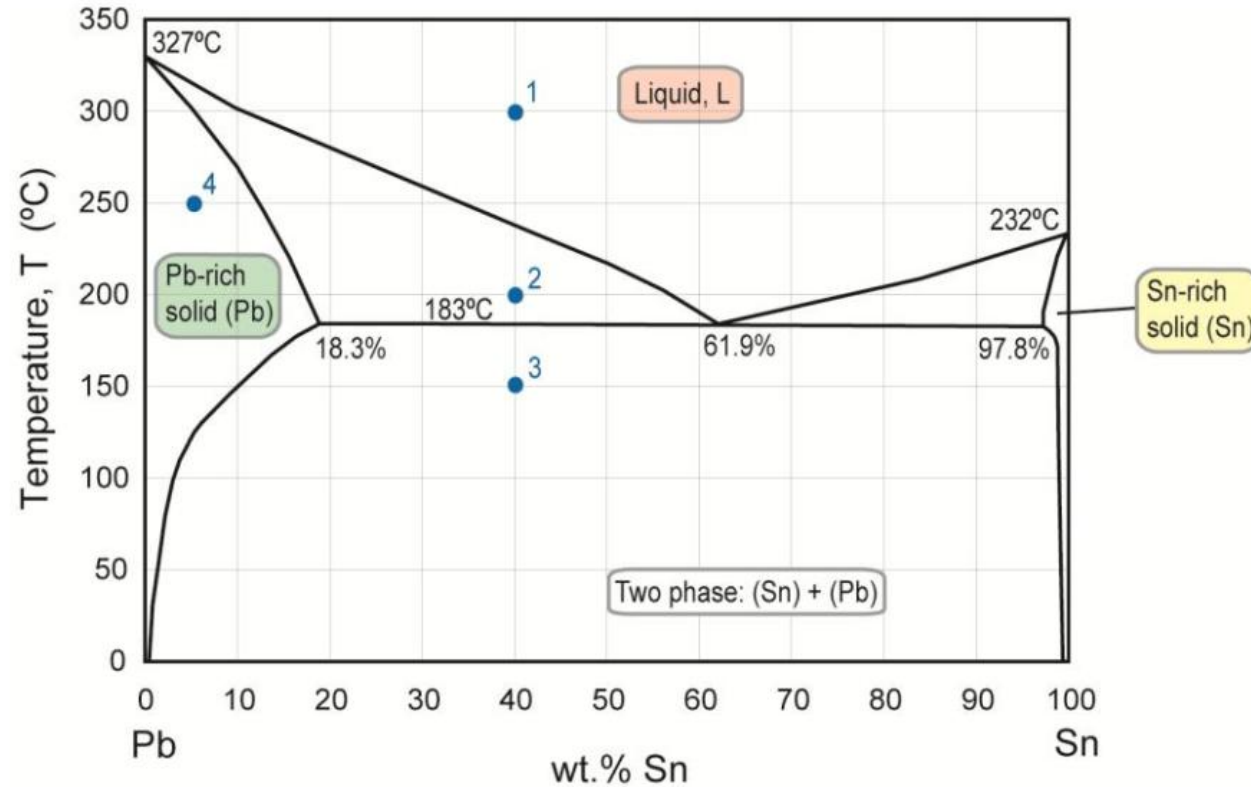




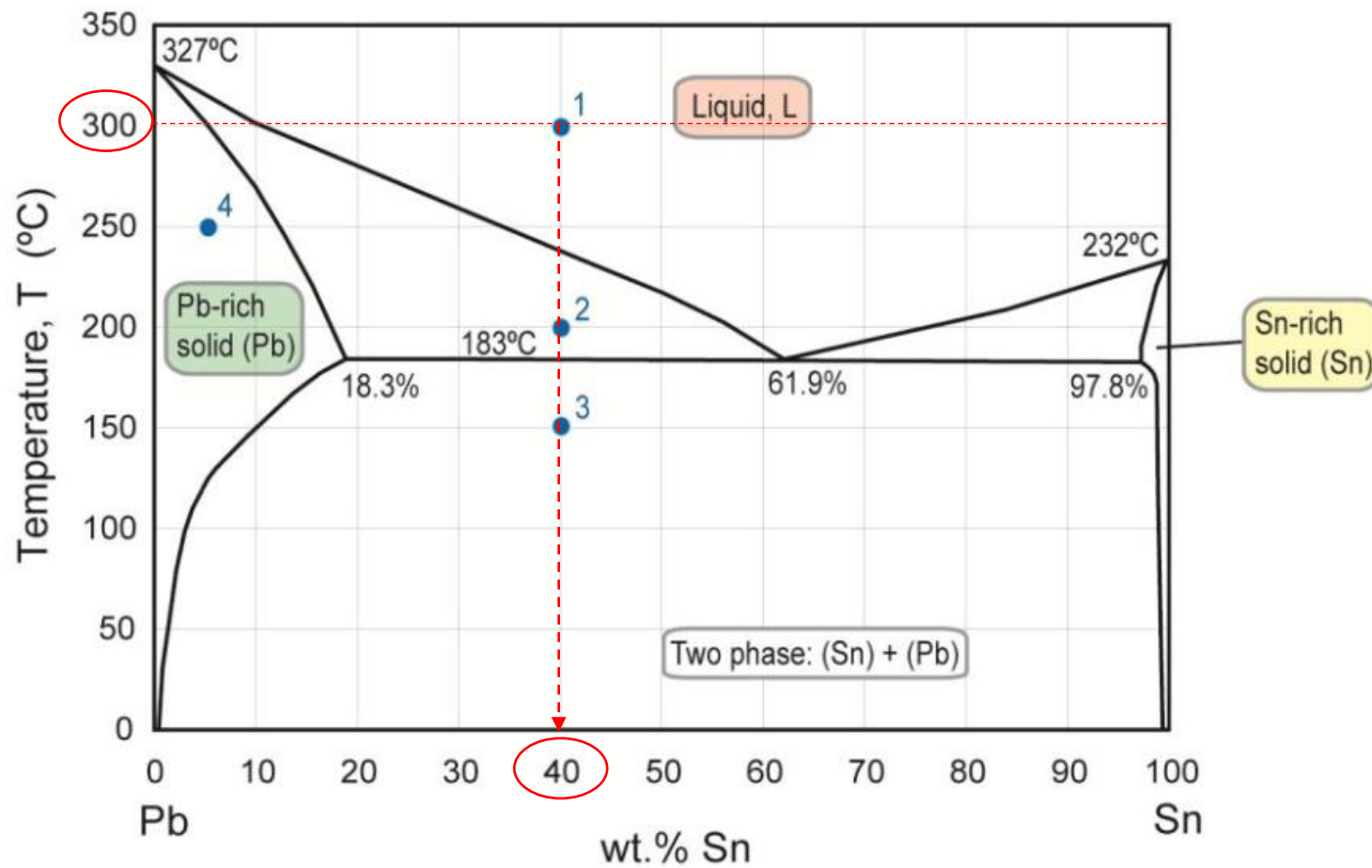
(F)	C	%
α	2% Sn 98% Pb	$\frac{99.5 - 10}{99.5 - 2} \cdot 100 = 92\%$
β	99.5% Sn 0.5% Pb	8%



E.6 Use the Pb-Sn diagram below to answer the following questions.



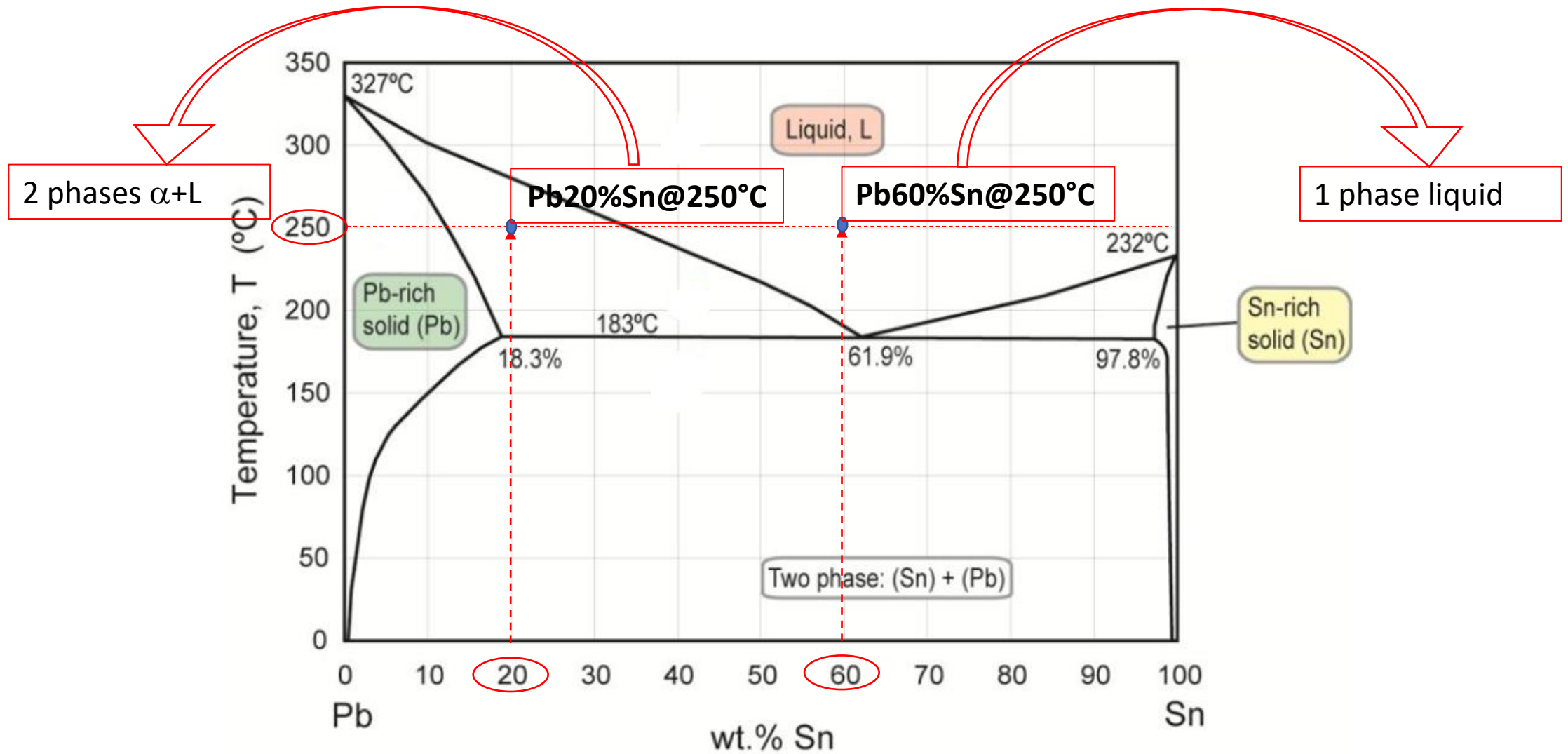
- (a) What are the values of the state variables (composition and temperature) at constitution point 1?
- (b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?
- (c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?
- (d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:
 - (i) the initial composition, temperature and phase(s);
 - (ii) the temperature at which a phase change occurs, and the final phase(s).



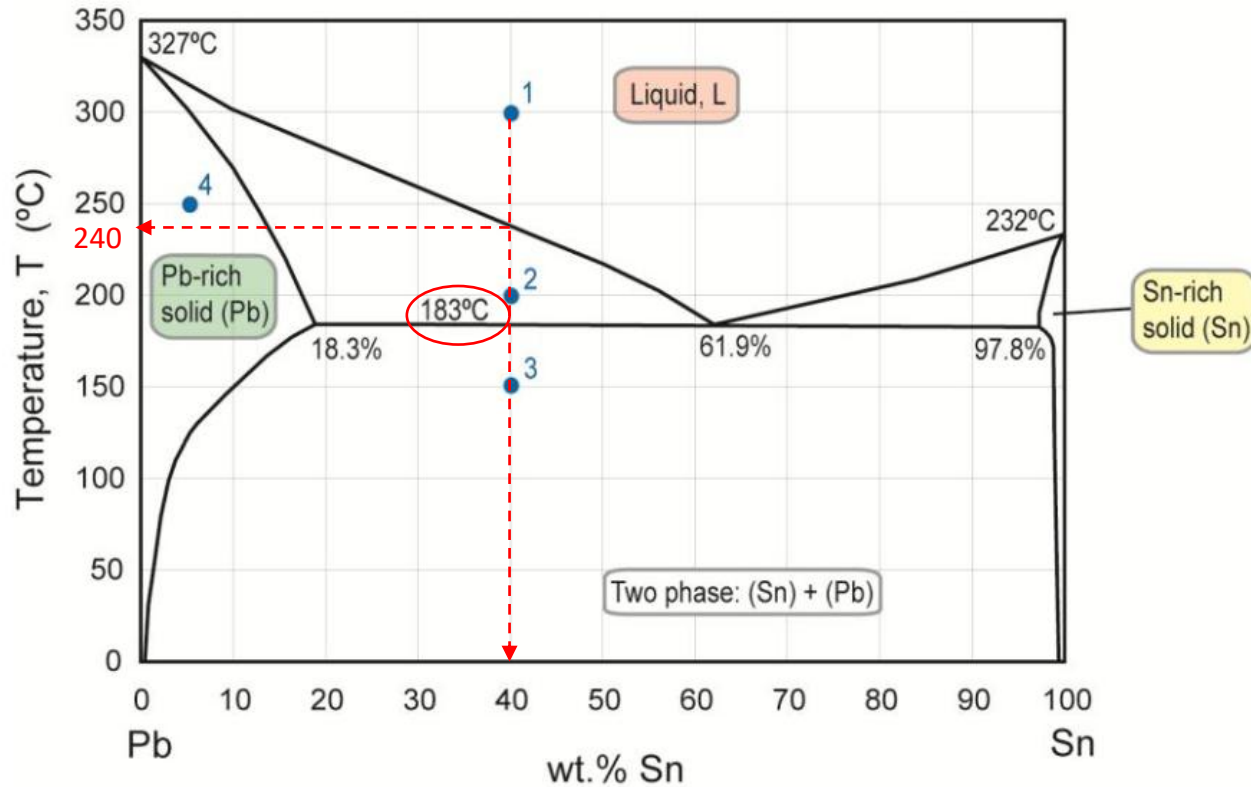
Point 1

1 phase liquid, composition
40%Sn, 60% Pb

Temperature 300 $^{\circ}\text{C}$



(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?



Alloy 1:

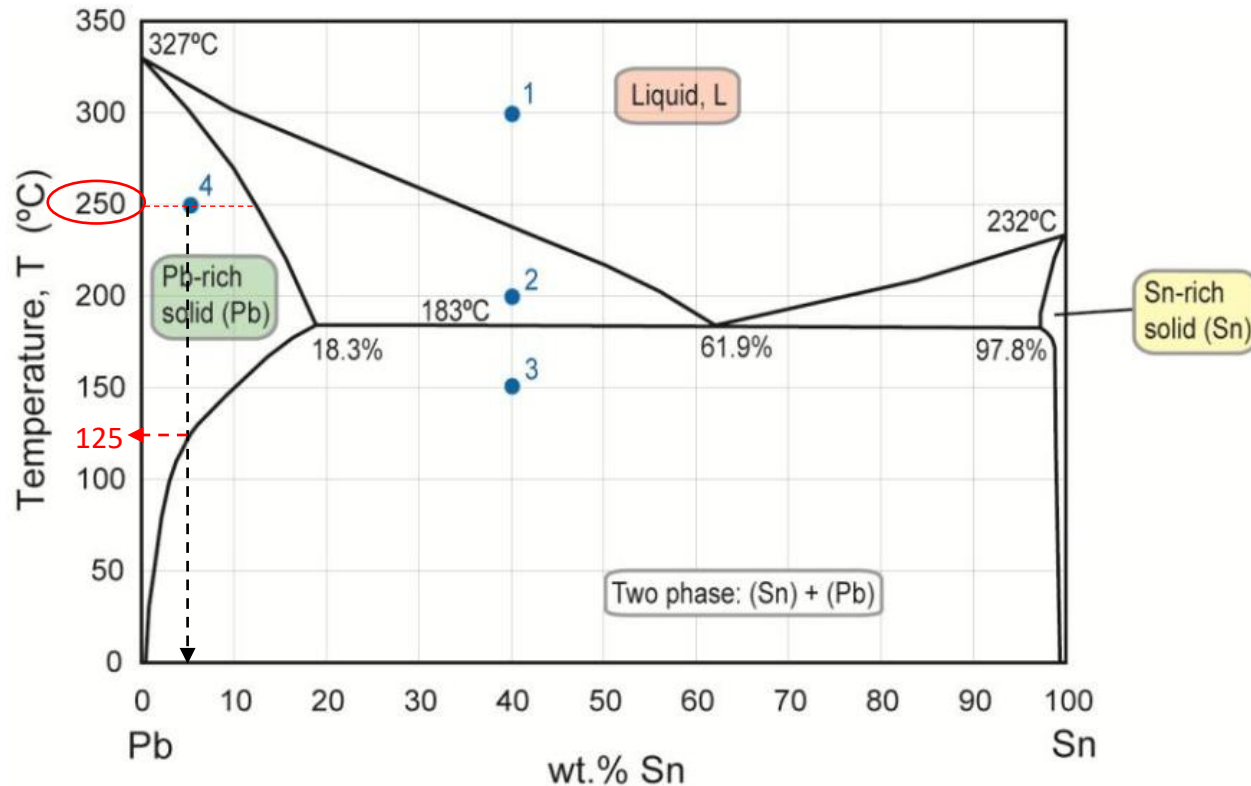
- 300°C – only liquid
- 240°C - solidification of the first α crystal
- 183°C - all the liquid undergoes eutectic transformation producing α - β lamellar microstructure

Point 2: 2 phases α (solid rich in Pb) and Liquid (L)

Point 3: 2 phases α (solid rich in Pb) and β (solid rich in Sn)

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:

- (i) the initial composition, temperature and phase(s);
- (ii) the temperature at which a phase change occurs, and the final phase(s).

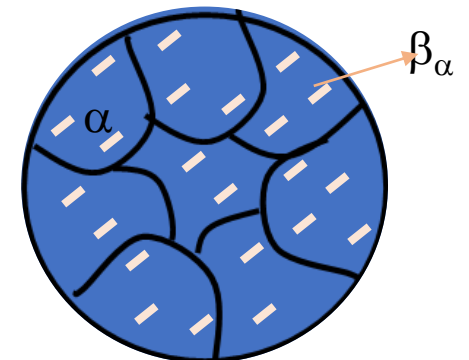


Alloy 4:

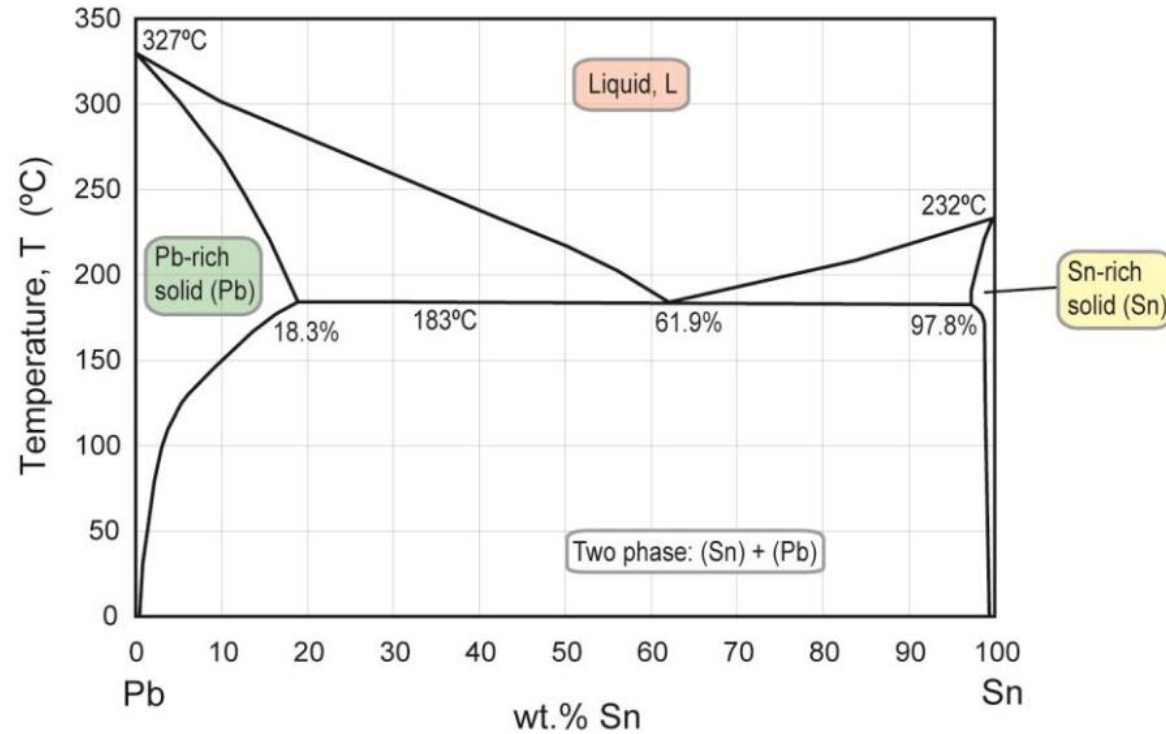
Start – 250°C, α (solid phase rich in Pb),
composition : 5%Sn, 95%Pb

125°C precipitation of β_{α} solid phase (rich in Sn)
due to reduction of the solubility of Sn in Pb at that
temperature

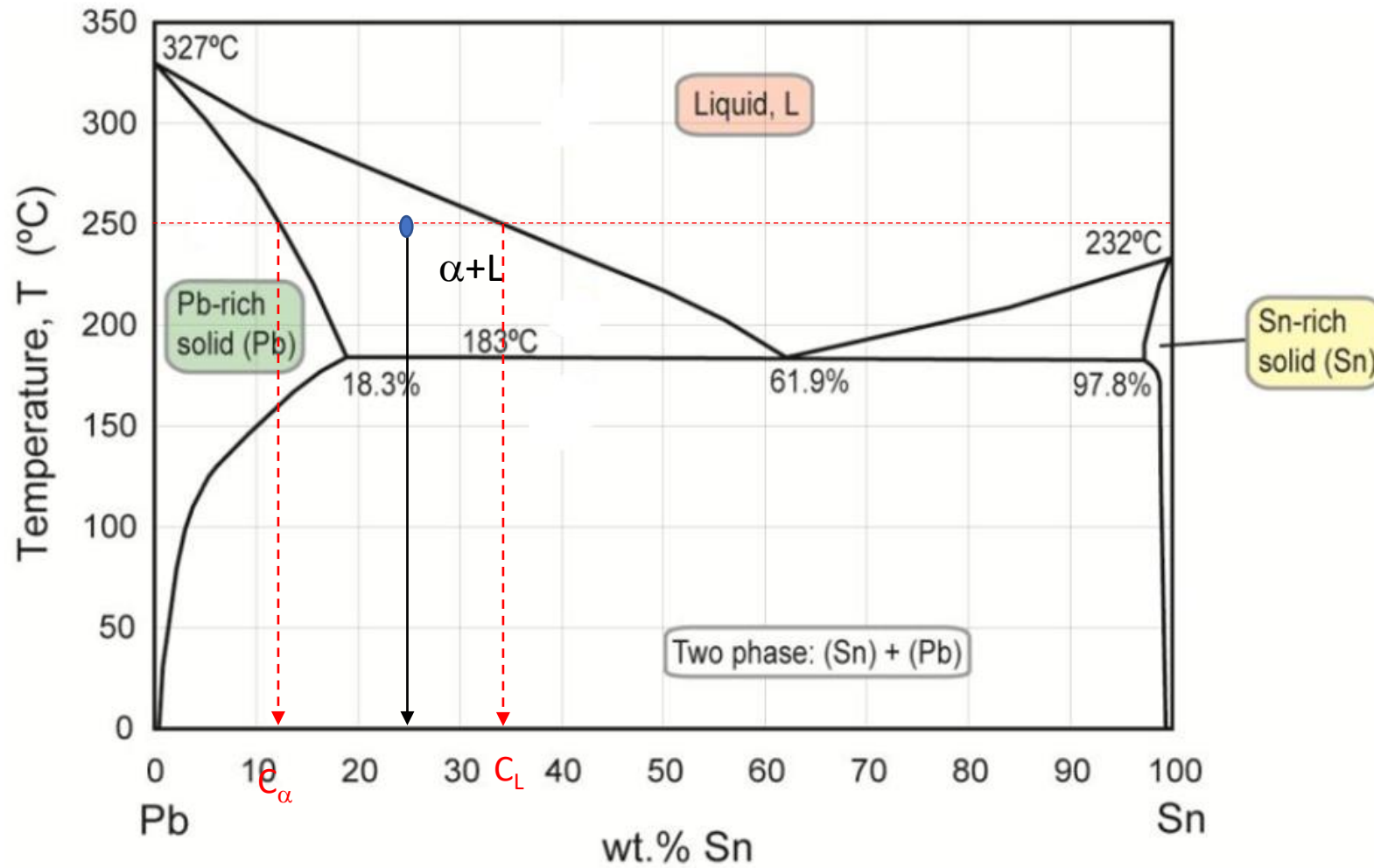
From 125°C to room temperature no other phase
changes occur, the microstructure is solid phase α
(rich in Pb) with small β_{α} crystals



E.7 Use the Pb-Sn diagram below to answer the following questions.



- (a) The constitution point for a Pb-25wt% Sn alloy at 250 °C lies in a two-phase field. Construct a tie-line on the figure and read off the two phases and their compositions.
- (b) The alloy is slowly cooled to 200 °C. At this temperature, identify the phases and their compositions
- (c) The alloy is cooled further to 150 °C. At this temperature, identify the phases and their compositions.
- (d) Indicate with arrows on the figure the lines along which the compositions of phase 1 and phase 2 move during slow cooling from 250°C to 200°C.

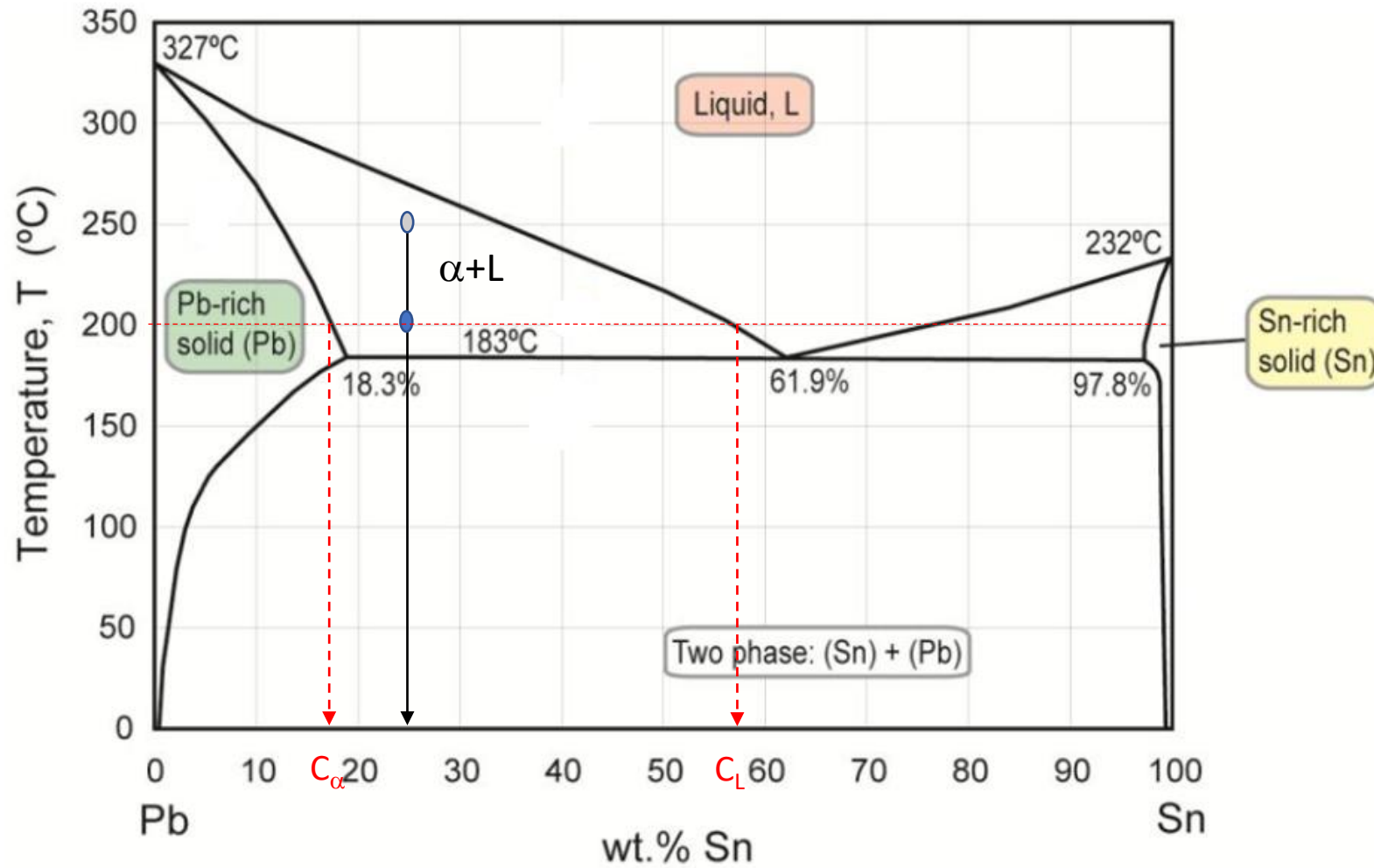


Alloy 25%Sn 75%Pb @ 250°C

α (solid phase rich in Pb) and liquid (L)

Composition of α =12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

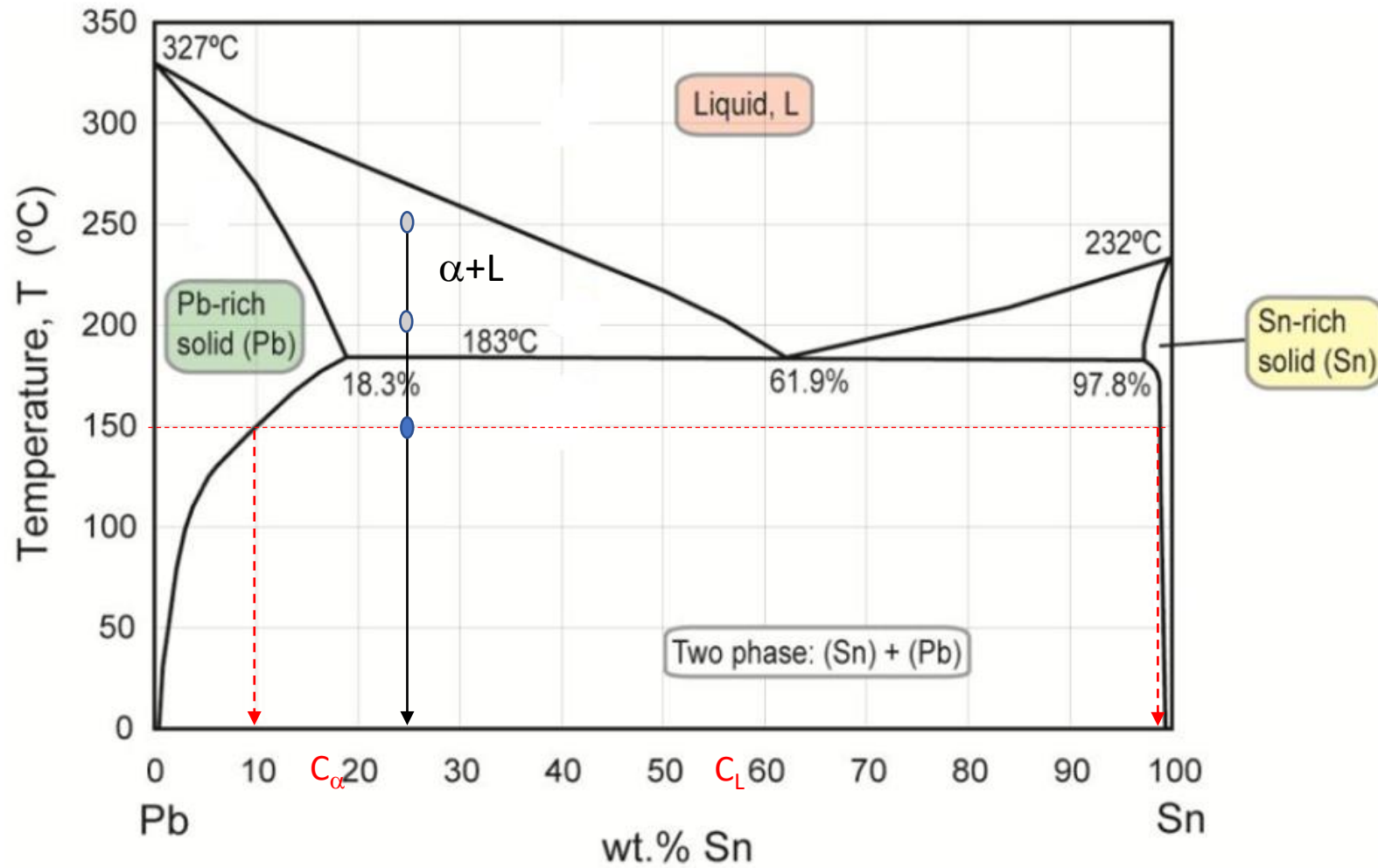


Alloy 25%Sn 75%Pb @ 200°C

α (solid phase rich in Pb) and liquid (L)

Composition of α =18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

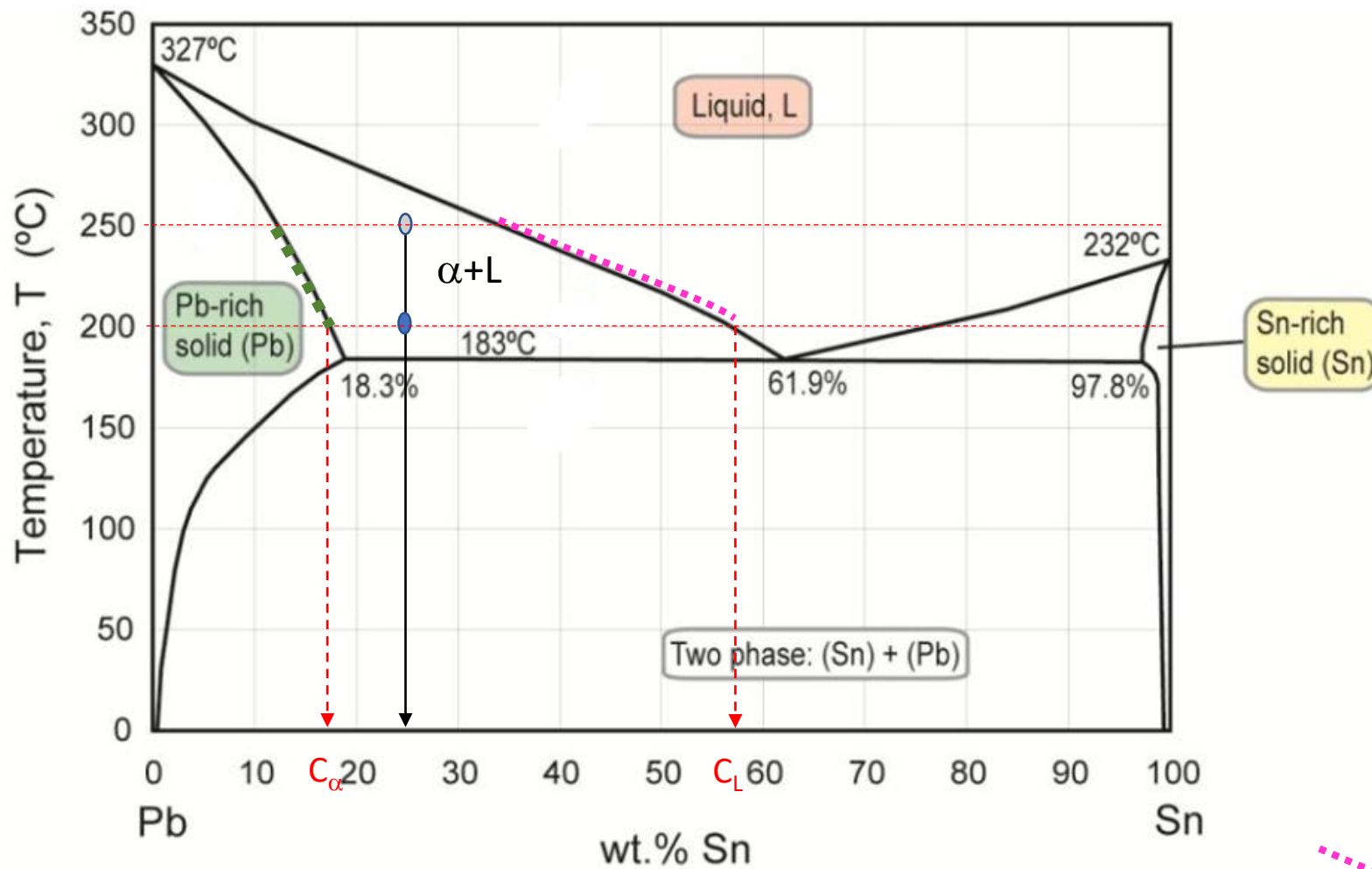


Alloy 25%Sn 75%Pb @ 150 $^{\circ}\text{C}$

α (solid phase rich in Pb) and
 β (solid rich in Sn)

Composition of α : 10%Sn,
 90%Pb

Composition of β : 99%Sn,
 1%Pb



Alloy 25%Sn 75%Pb @ 250°C

α (solid phase rich in Pb) and liquid (L)

Composition of α = 12%Sn, 88%Pb

Composition of liquid: 33%Sn, 67%Pb

Alloy 25%Sn 75%Pb @ 200°C

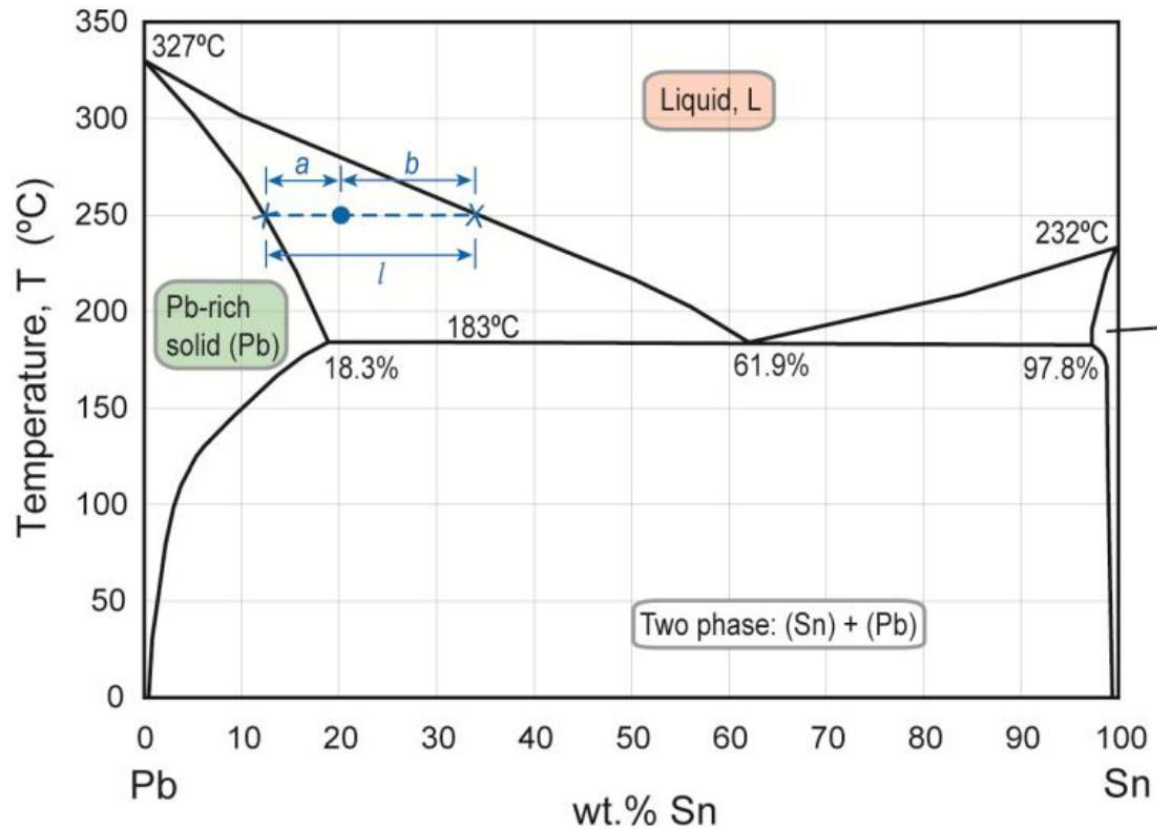
α (solid phase rich in Pb) and liquid (L)

Composition of α = 18%Sn, 82%Pb

Composition of liquid: 58%Sn, 42%Pb

Variation of α composition reducing temperature from 250 to 200 °C

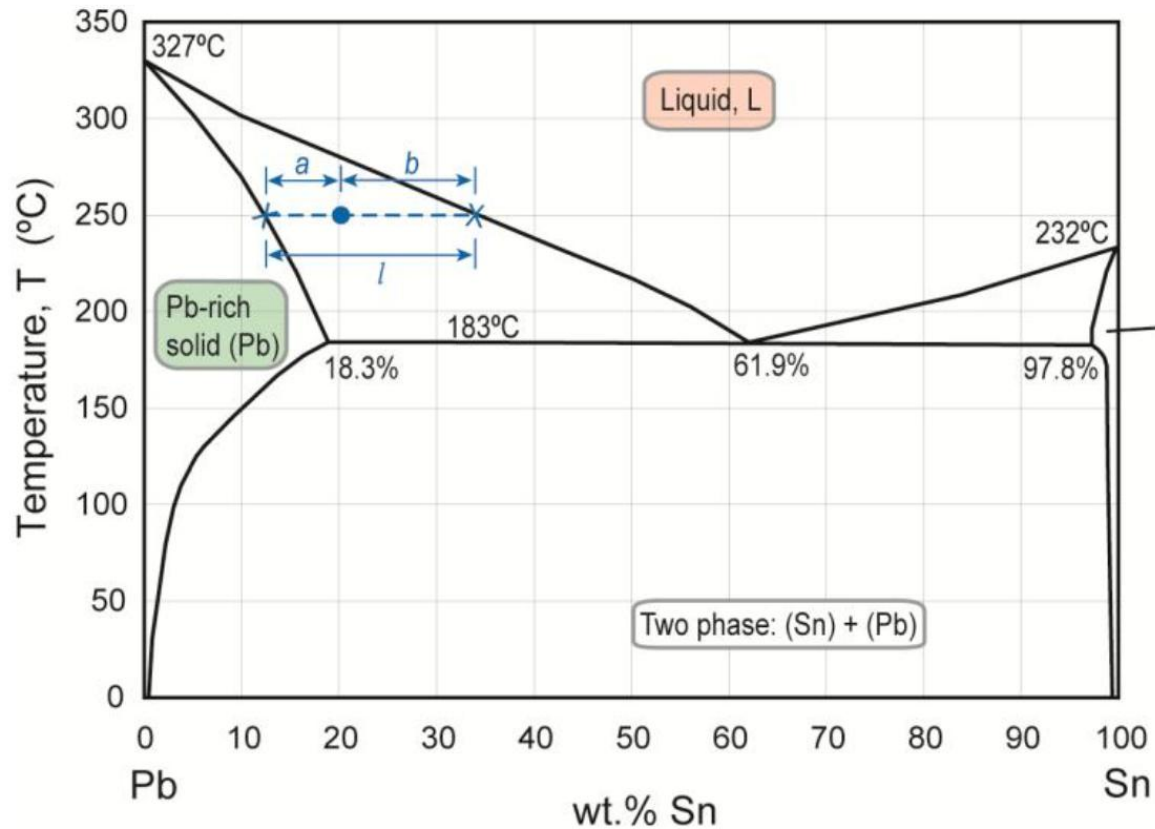
Variation of L composition reducing temperature from 250 to 200 °C



Consider Pb-20wt% Sn at 250 $^{\circ}\text{C}$ in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

Find the proportions of each phase



Consider Pb-20wt% Sn at 250 °C in Figure

A constitution point in the two-phase field: liquid plus Pb-rich solid.

To find the proportions of each phase

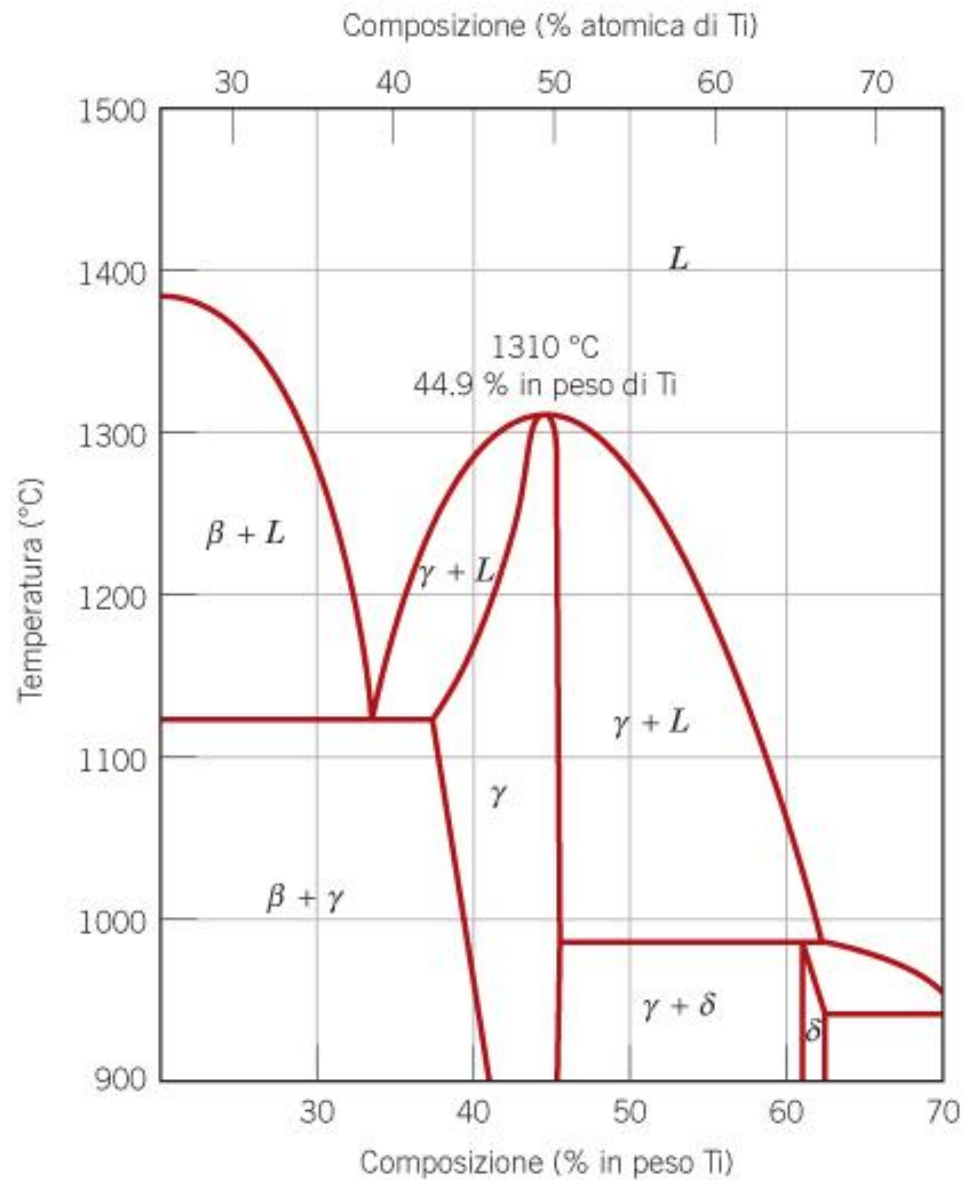
First construct a tie-line through the constitution point and read off the compositions of the phases:

Pb-rich solid with $W_{Sn}^{SOL} = 12\%$; *Liquid* with $W_{Sn}^{LIQ} = 34\%$.

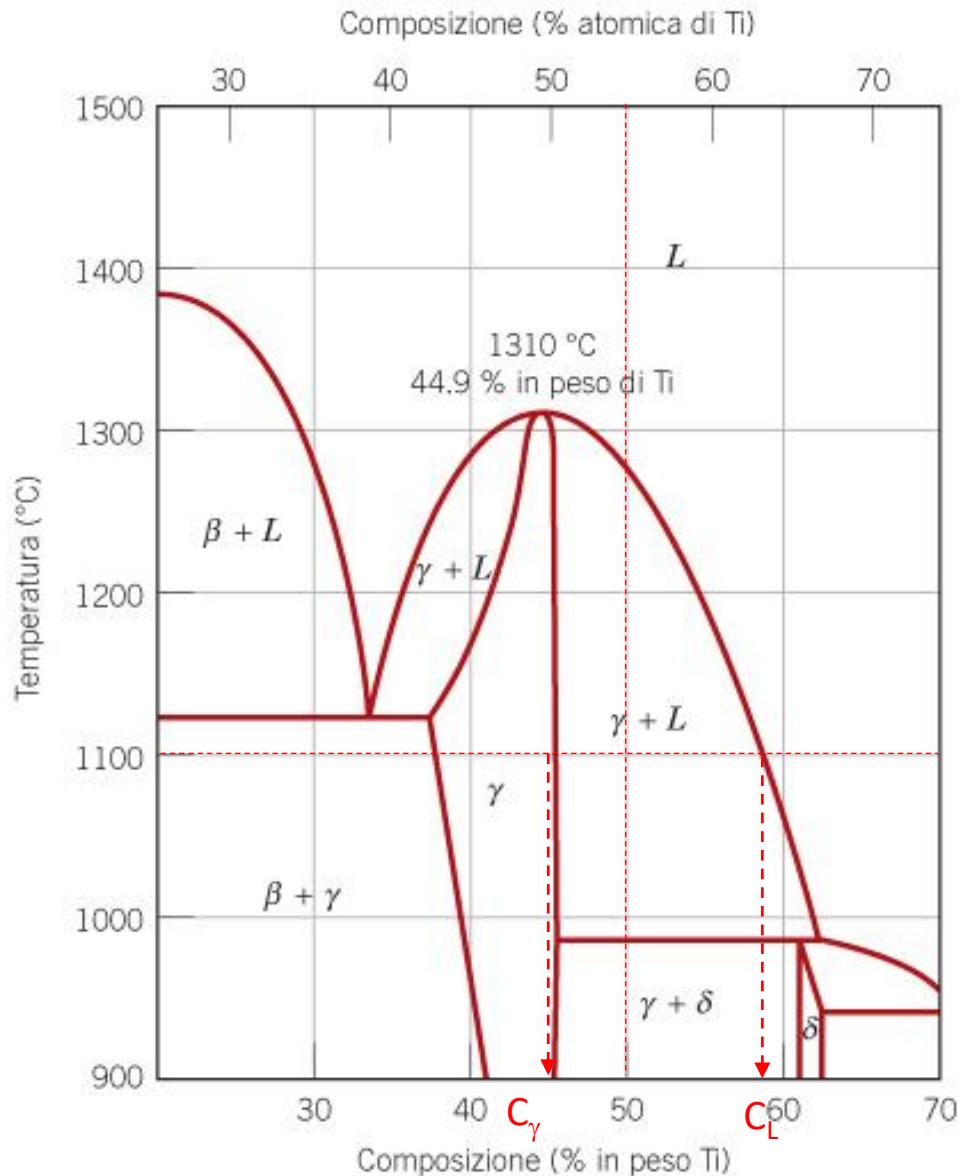
The tie-line is of length l , while the lengths of the segments to either side of the constitution point are **a** and **b** respectively

For the example alloy of composition $W_{Sn} = 20\%$

Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.



Phases: solid phase γ (intermediate solid solution) and liquid L

Compositions:

Composition of γ : 45%Ti

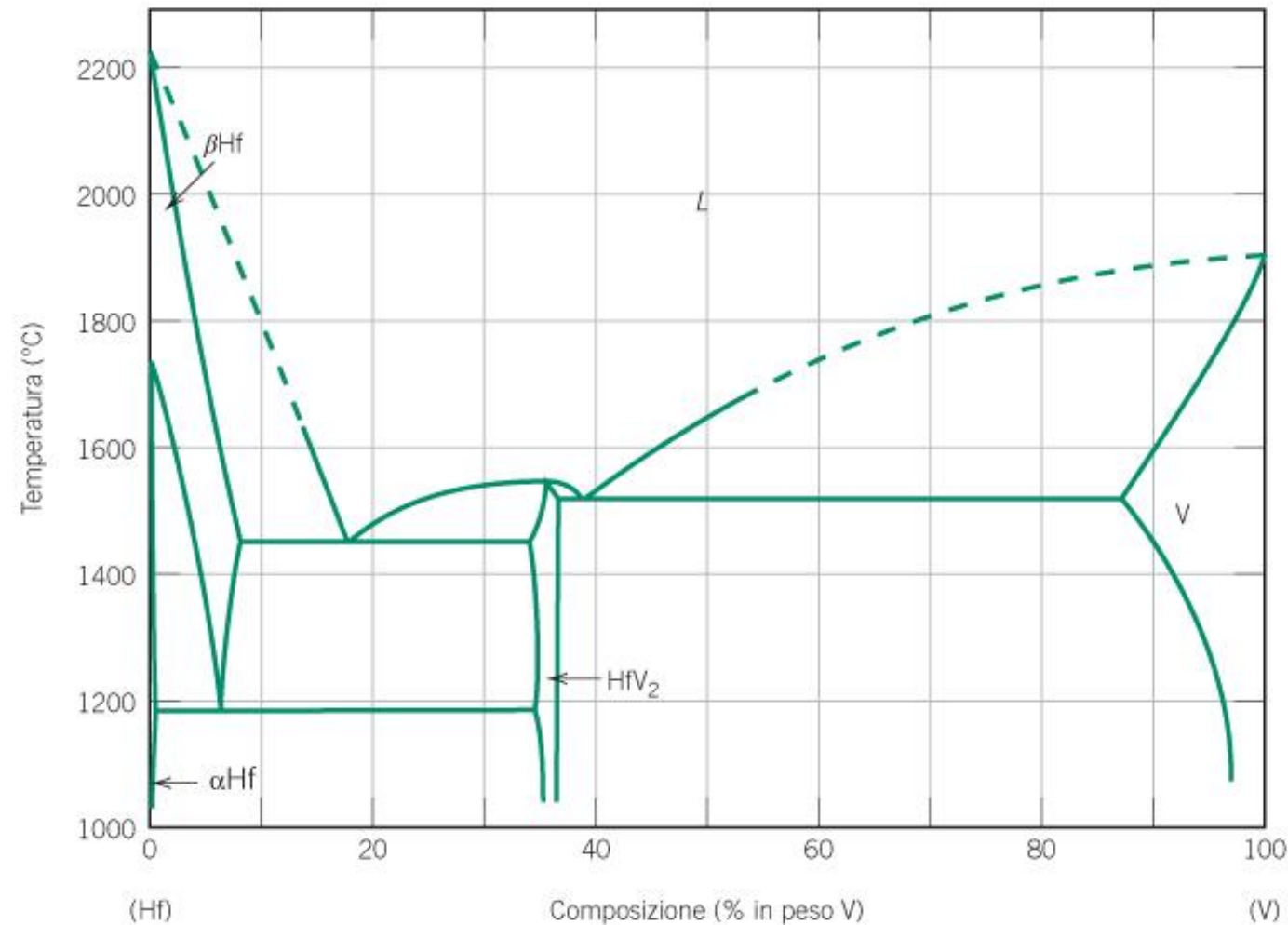
Composition of Liquid: 58%Ti

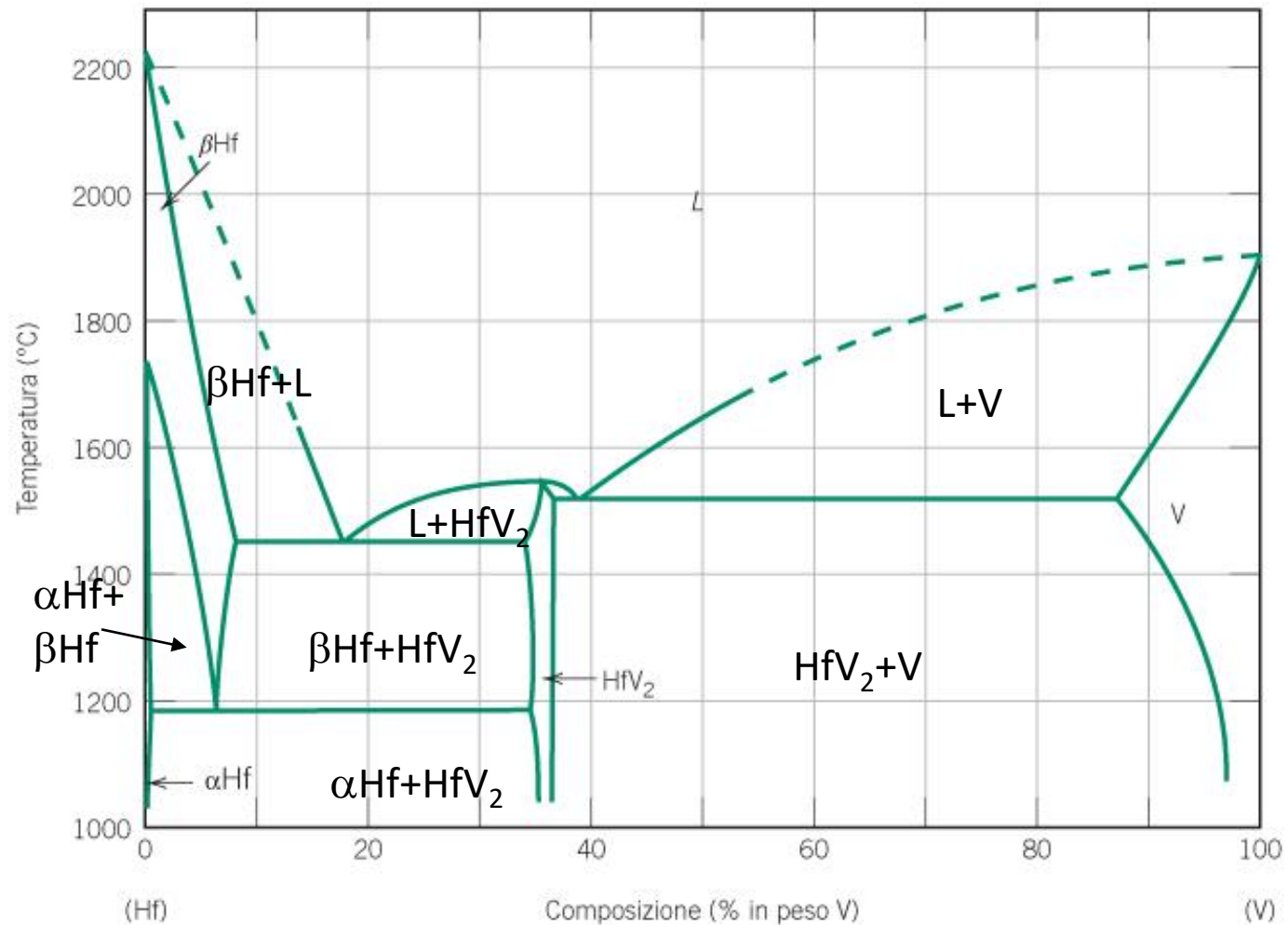
Relative amounts

$$\% \gamma = (58 - 50) / (58 - 45) * 100 = 61.5\%$$

$$\% L = (50 - 45) / (58 - 45) * 100 = 38.5\%$$

Complete the following diagram





Complete the following diagram

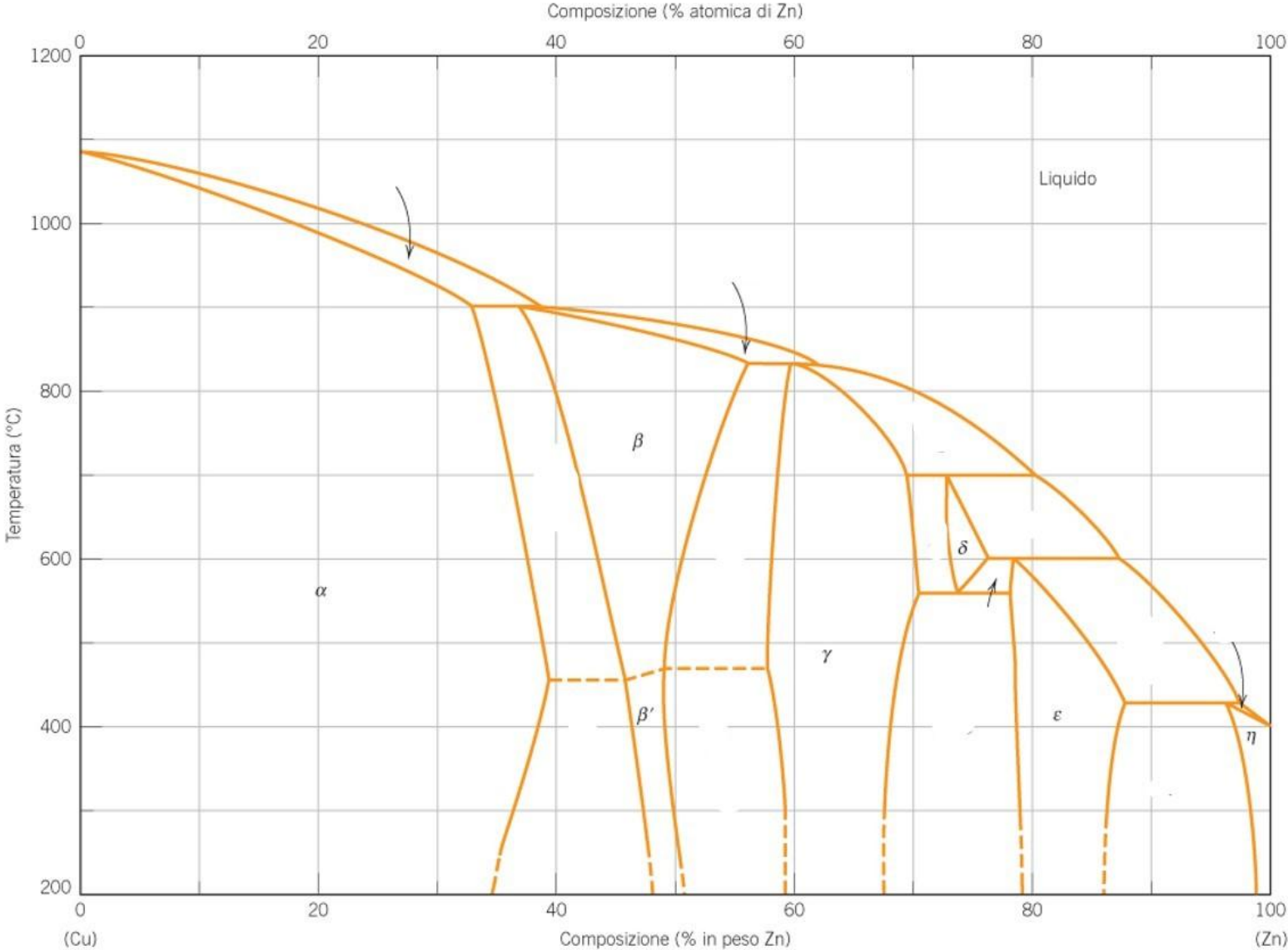


Figura 9.19 Diagramma di fase rame-zinco.



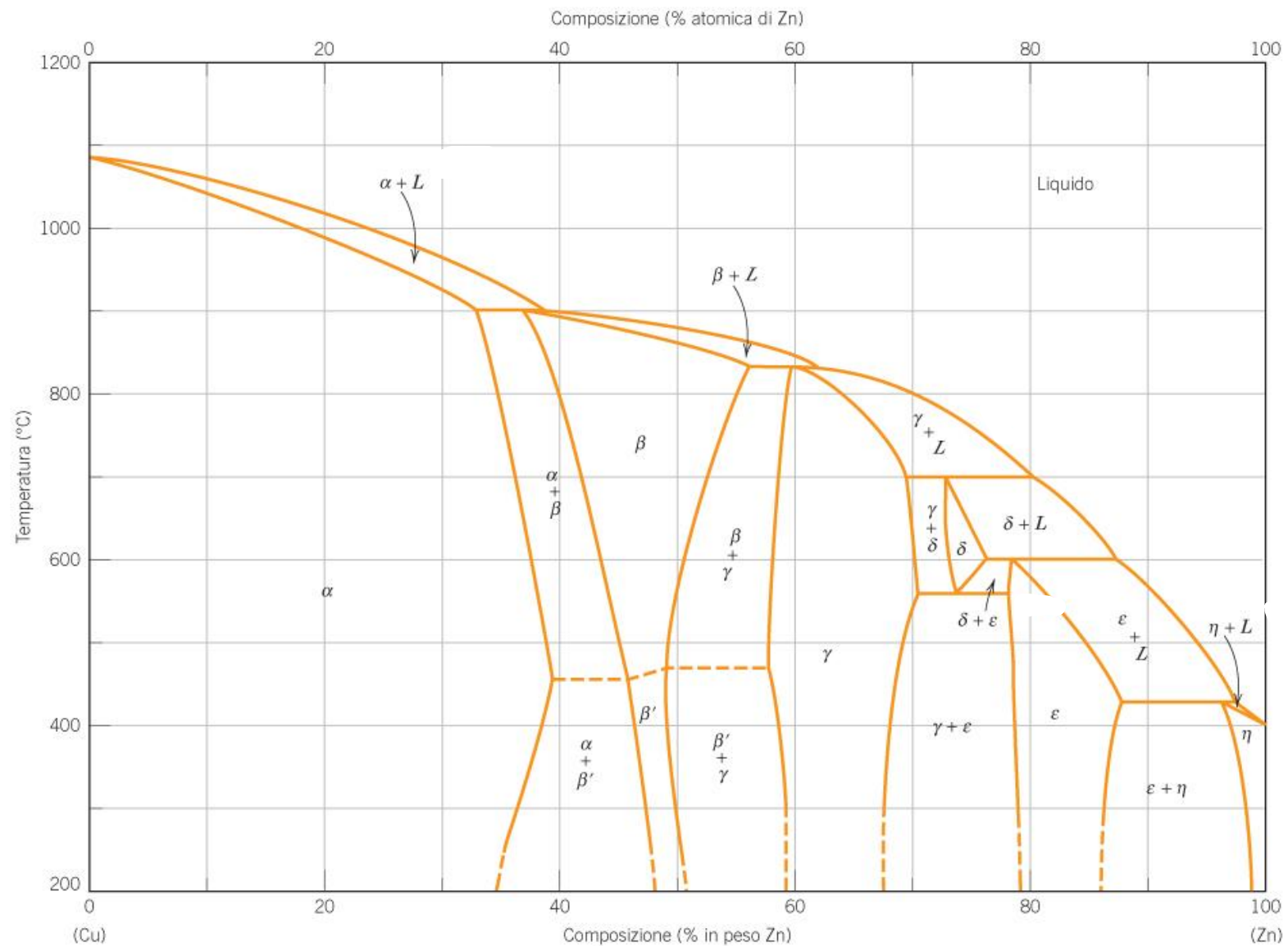


Figura 9.19 Diagramma di fase rame-zinco.